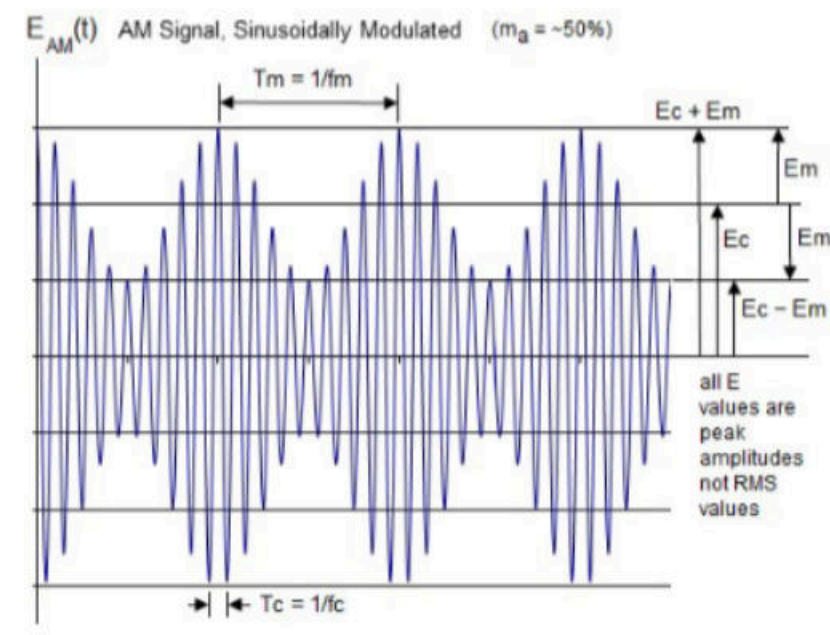


Sarah Boyd
Group members: Wesley Hwee and Mariana Rivera
ETE 3351
Lab 7

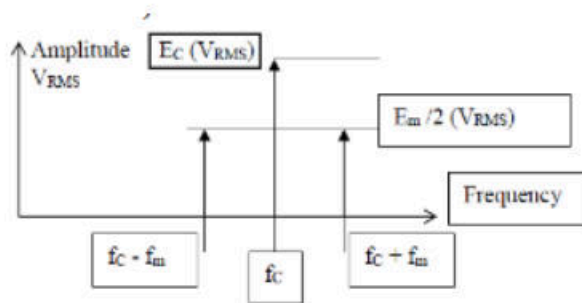
AM Laboratory

Introduction

This lab introduces amplitude modulation (AM) and shows how an AM signal can be understood by looking at both the time domain and the frequency domain. In amplitude modulation a high-frequency carrier signal is used to transmit information by changing its amplitude based on a lower-frequency message signal. On an oscilloscope this appears as a fast carrier wave surrounded by a slower-changing envelope, while in the frequency domain the same signal appears as a carrier frequency with additional sideband frequencies above and below it. By using both the oscilloscope and the spectrum analyzer the relationship between the carrier amplitude and frequency, the modulation amplitude and frequency and the modulation index can be clearly observed.



IMG 1. Time domain



IMG 2. *Frequency domain*

The time domain equation for a sinusoidally modulated AM signal is $E_{AM}(t) = (E_c + E_m \cos 2\pi f_m t) \cos 2\pi f_c t$ where

- E_c : the carrier amplitude (in v)
- E_m : the message amplitude (in v)
- F_c : the carrier frequency
- F_m : the message frequency
- M : the modulation index, $= \frac{E_m}{E_c}$, where

$m=0$ means no modulation (just the carrier)

$0 < m < 1$ means normal AM

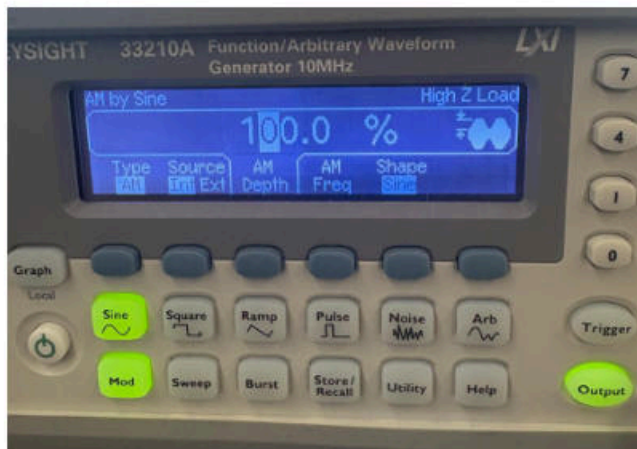
$m=1$ means 100% modulation

$m > 1$ over modulation, distortion

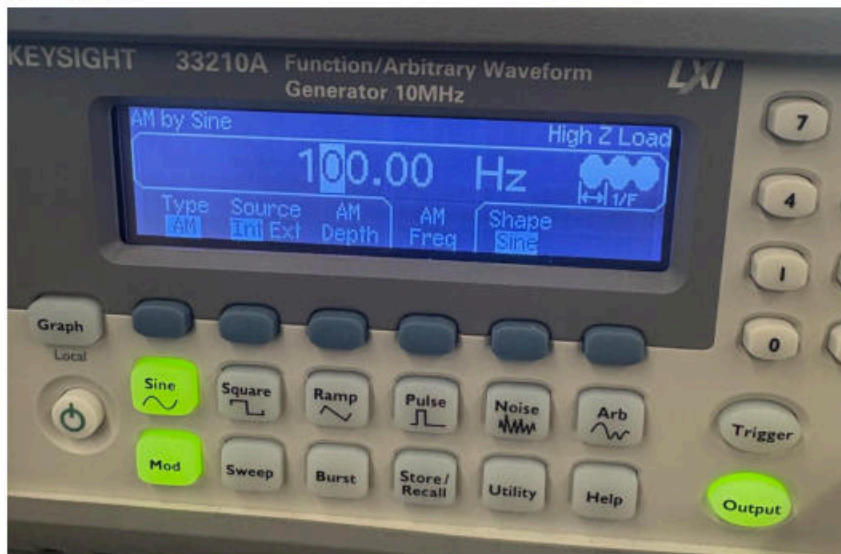
As the modulation index increases the envelope becomes more pronounced and over-modulation causes visible distortion in both the time and frequency domains. Non-sinusoidal modulation signals such as triangle and square waves are also examined in this lab. Finally, the lab demonstrates AM demodulation using a simple diode detector circuit which recovers the original message signal from the AM waveform. Overall, this lab helps build an intuitive understanding of how AM signals and am demodulation works.

In the lab:

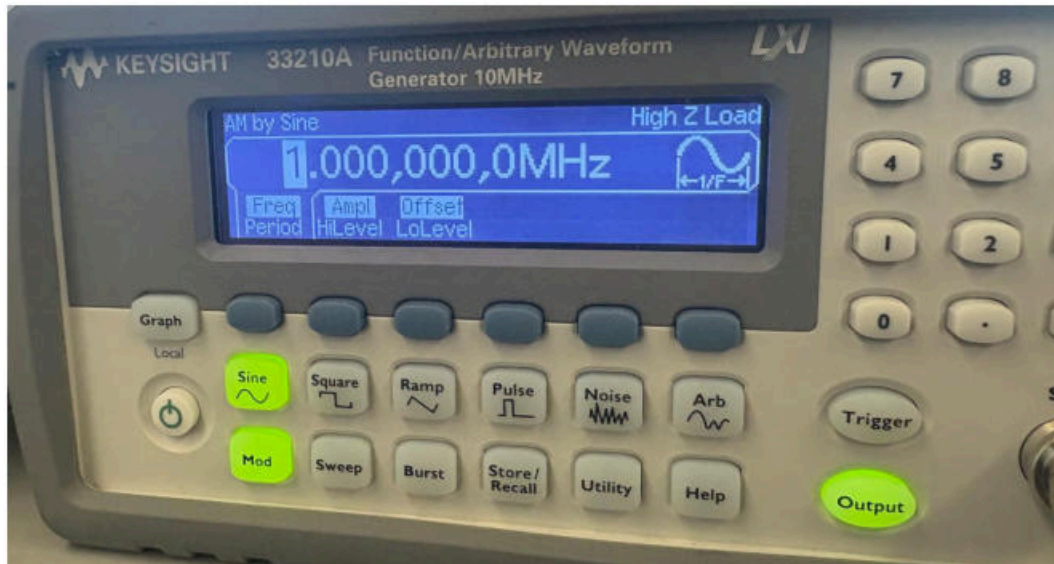
1. Take enough lab data and screenshots to confirm E_c , f_c , E_m , and f_m values in the time and frequency domains. Suggested starting values $f_c = 100 \text{ kHz}$ to 1 MHz , $E_c = 2 \text{ Volts}_{\text{pk-pk}}$, $f_m = 10 \text{ kHz}$, E_m values for m_a of 0, 0.25, 0.5, 0.75, and 1 (0%, 25%, 50%, 75%, and 100% modulation) where with $m_a = E_m/E_c \times 100\%$



IMG 3. m_a



IMG 4. f_m

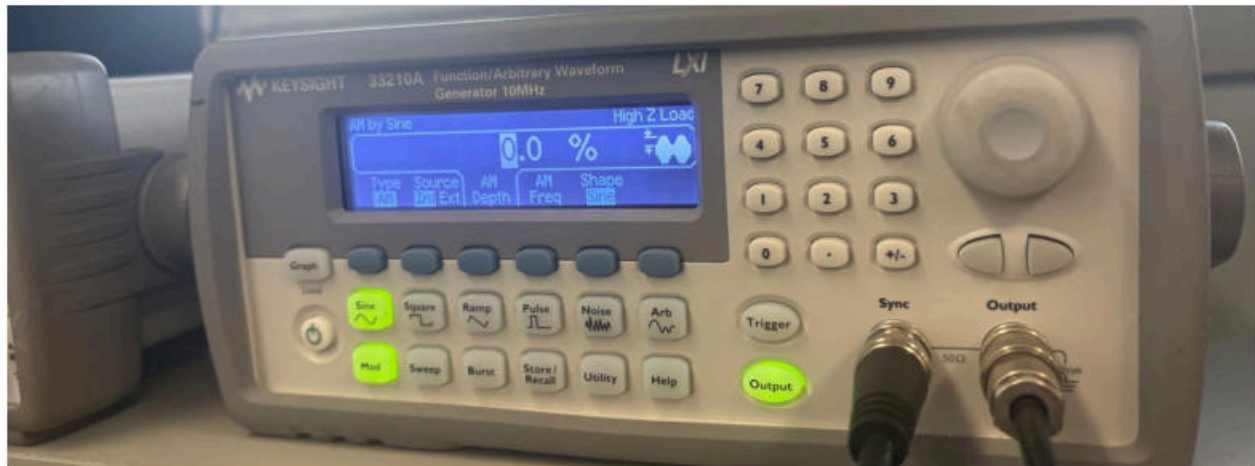


IMG 5.fc



IMG 6. The set up

0% modulation:



IMG 7.

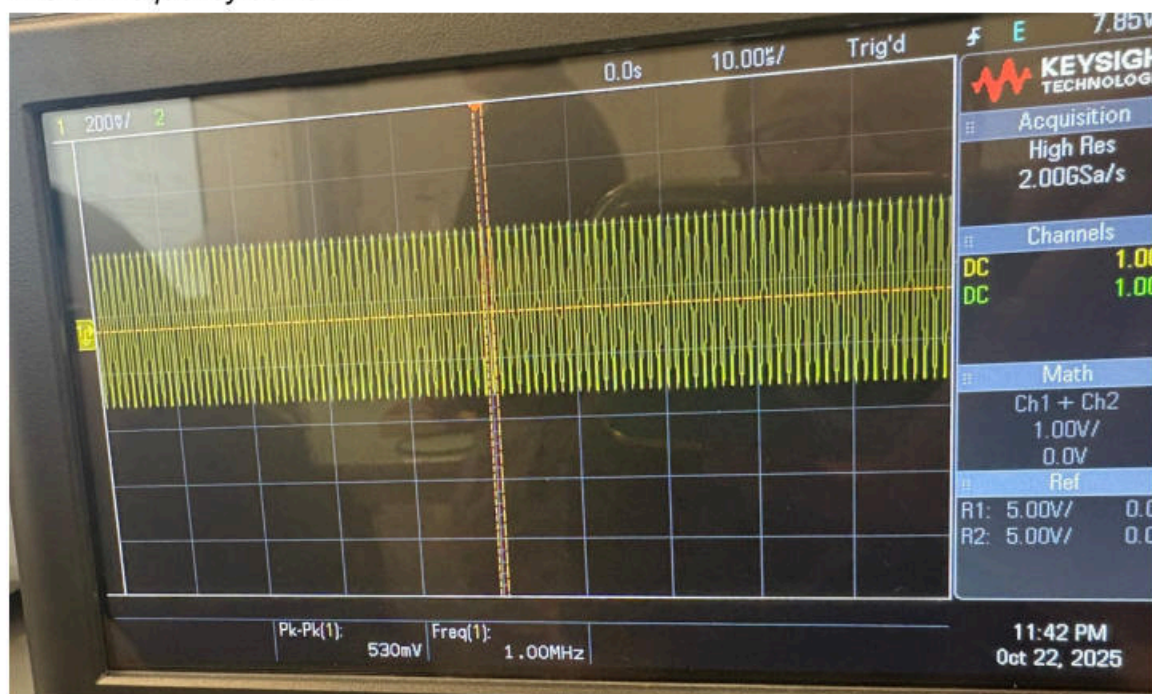


IMG 8. *Fm*

carrier:(Ec)



IMG 9. Frequency domain

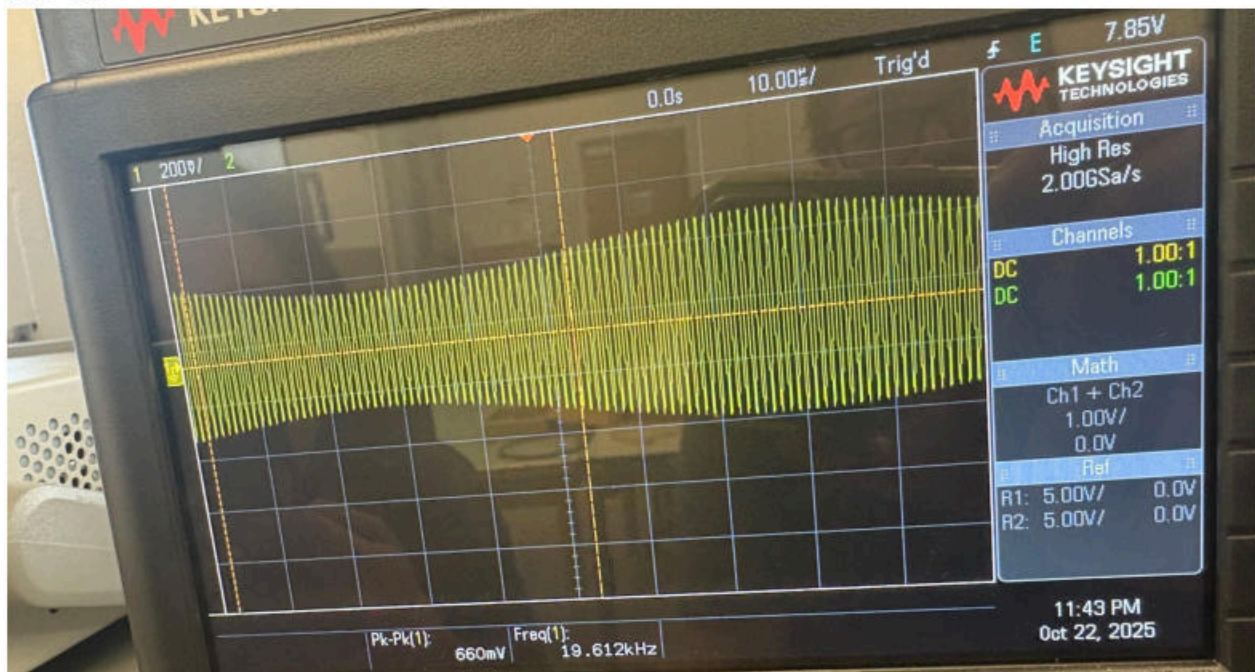


IMG 10. time domain

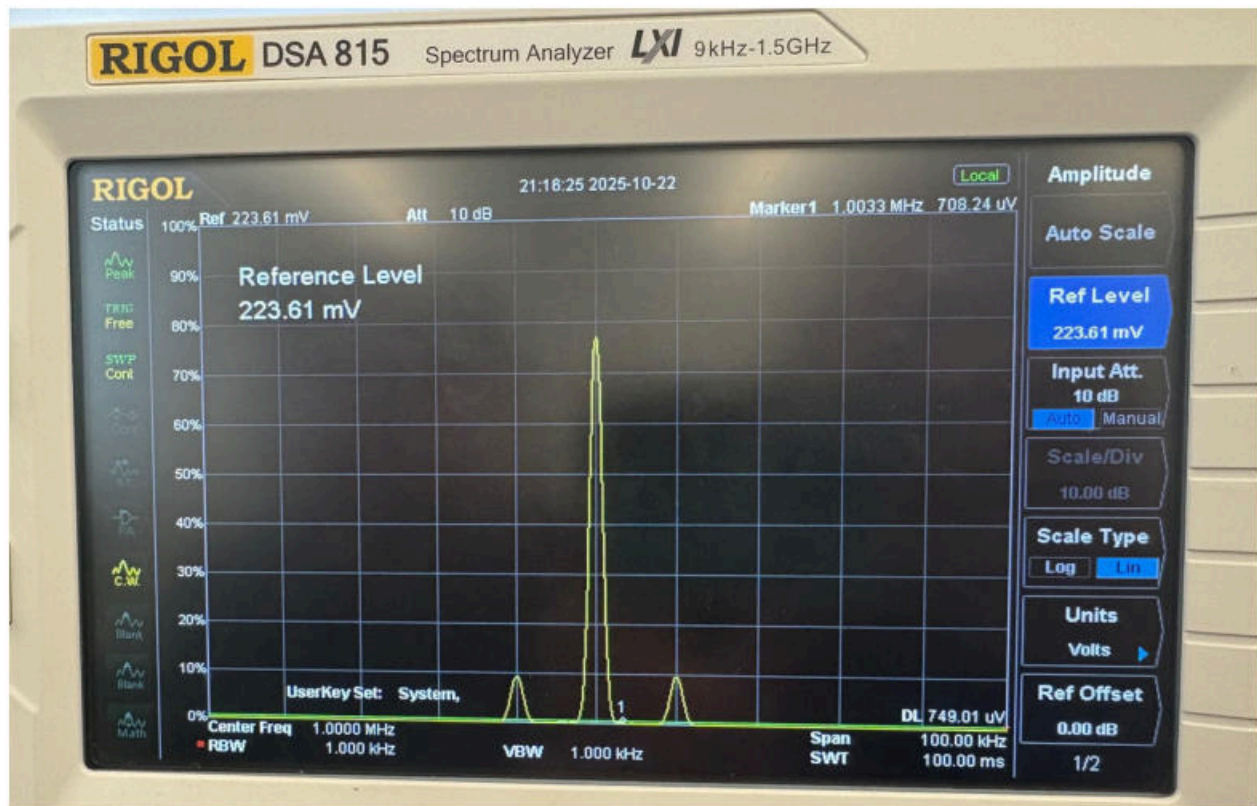
25% modulation:



IMG 11.

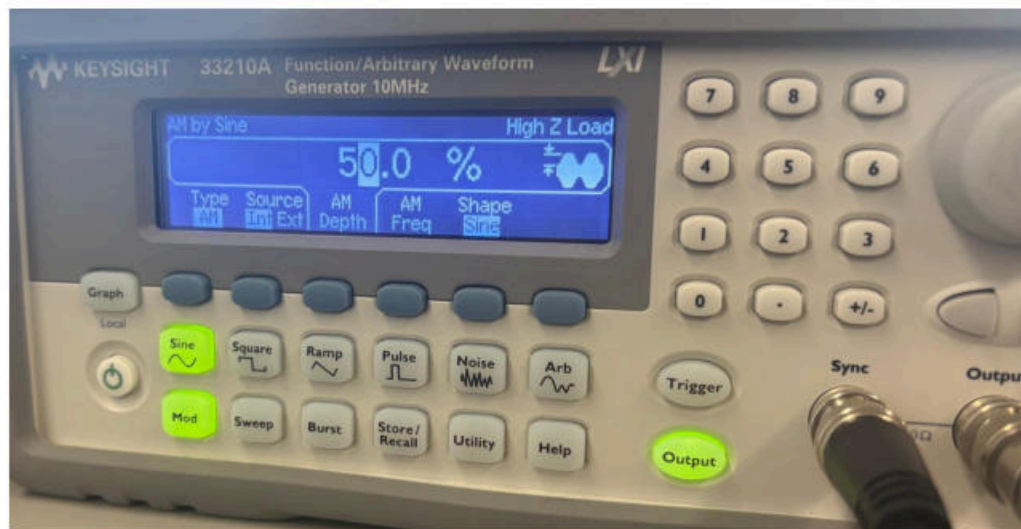


IMG 12. time domain

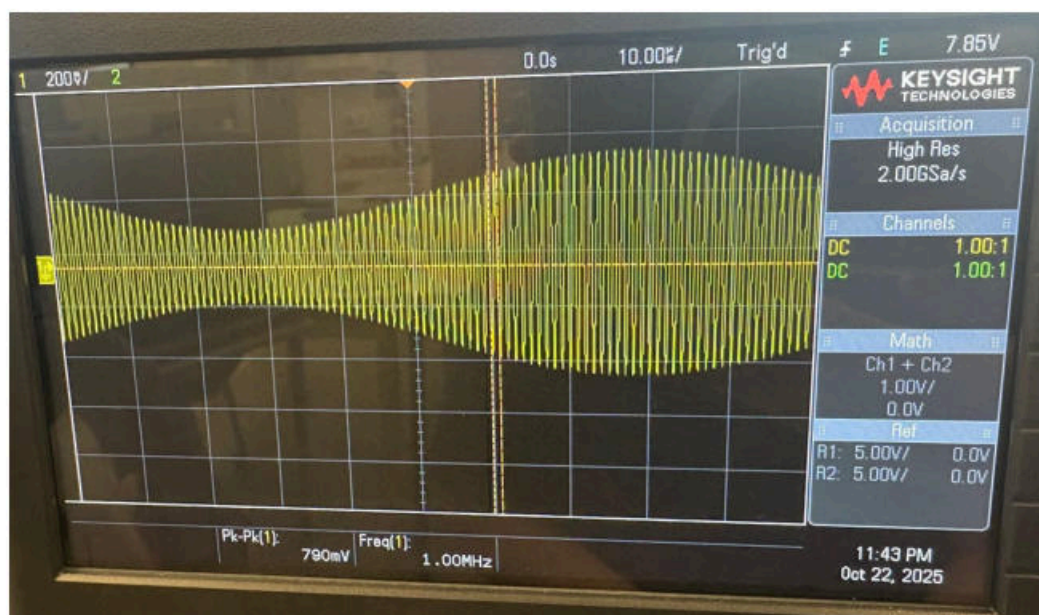


IMG 13. Frequency domain

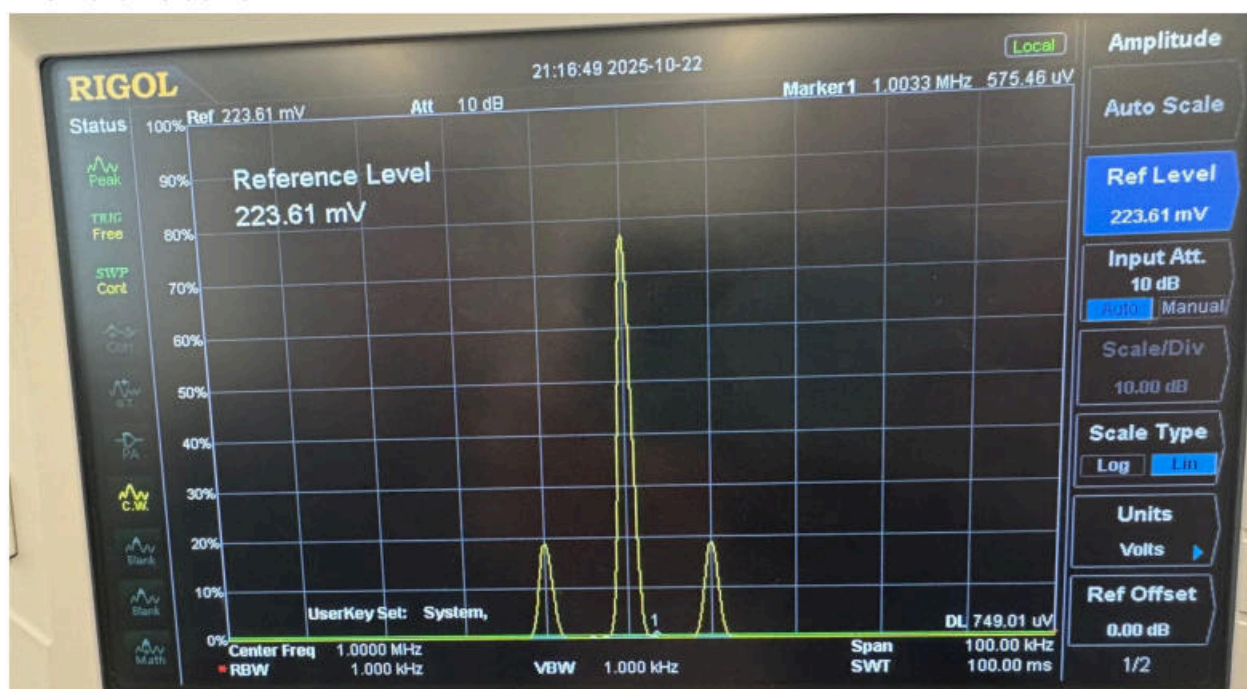
50% modulation:



IMG 14.



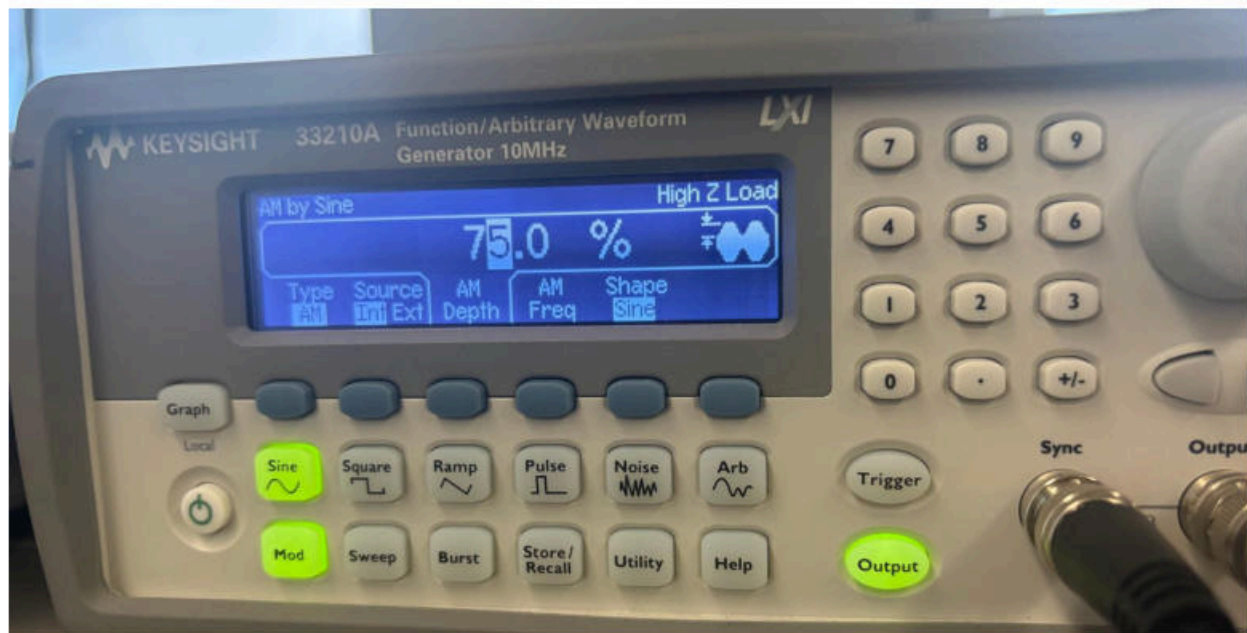
IMG 15. time domain



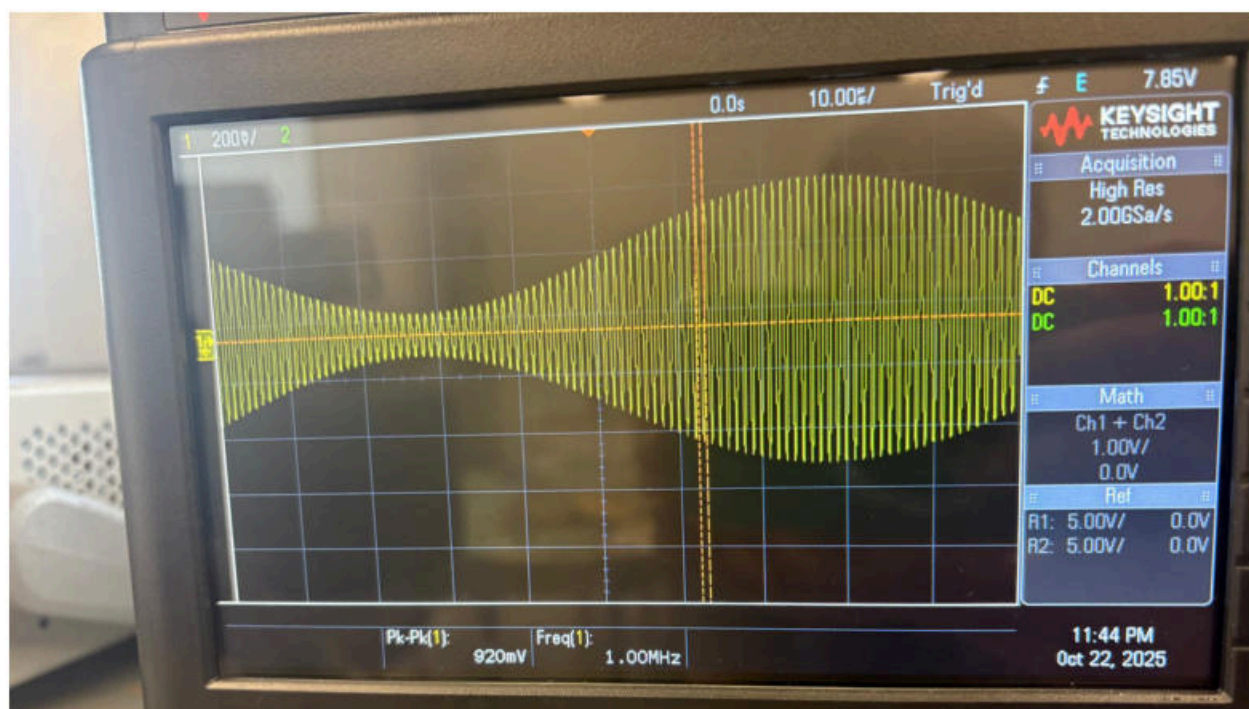
IMG 16. Frequency domain

What is going on? In this lab the modulation frequency f_m was kept constant at 10 kHz and the carrier frequency f_c was set to 1 MHz. The carrier amplitude E_c represents the voltage of the unmodulated carrier and is observed when the modulation index is set to 0 percent (where only the carrier is present and the signal has a constant amplitude). When modulation is applied E_m is at the voltage amplitude based on E_c . As the modulation index increases from 25 percent to 50 percent, 75 percent, and finally 100 percent each oscilloscope image shows a deeper envelope while the carrier frequency itself remains unchanged. These changes directly represent the modulation term in the AM signal, where the carrier amplitude is varied by the modulating signal over time. Each captured image therefore demonstrates how the message signal controls the carrier's amplitude while the carrier frequency stays fixed. This confirms the relationship between the observed waveforms and the AM signal equation showing that only the amplitude of the carrier changes and not its frequency.

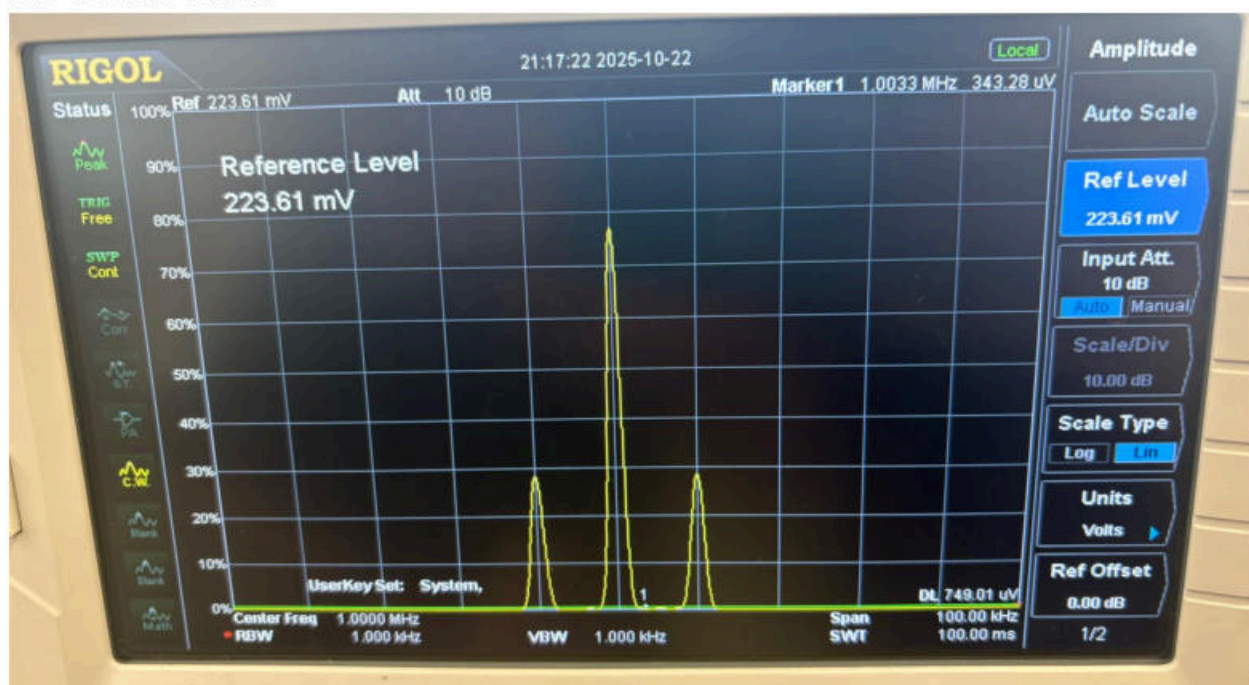
75% modulation:



IMG 17

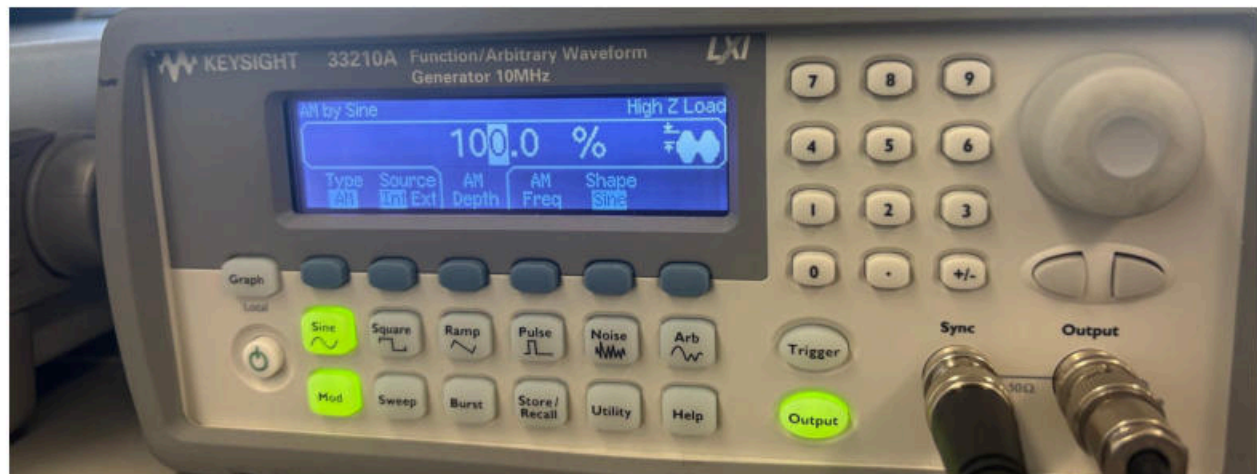


IMG 18. time domain

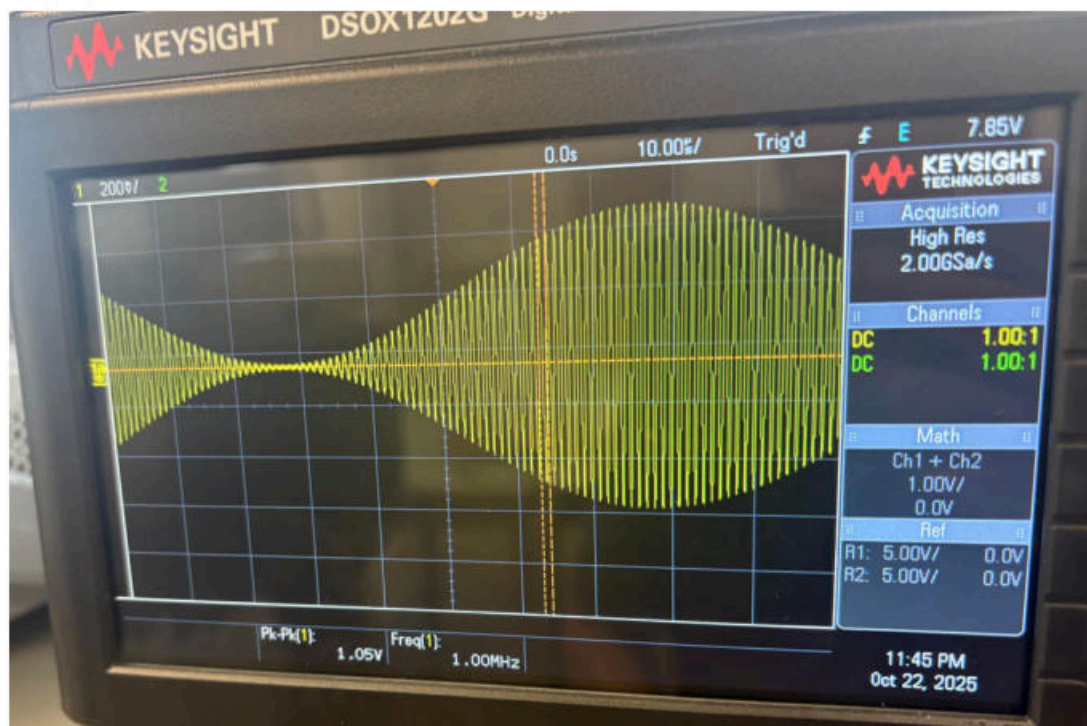


IMG 19. Frequency domain

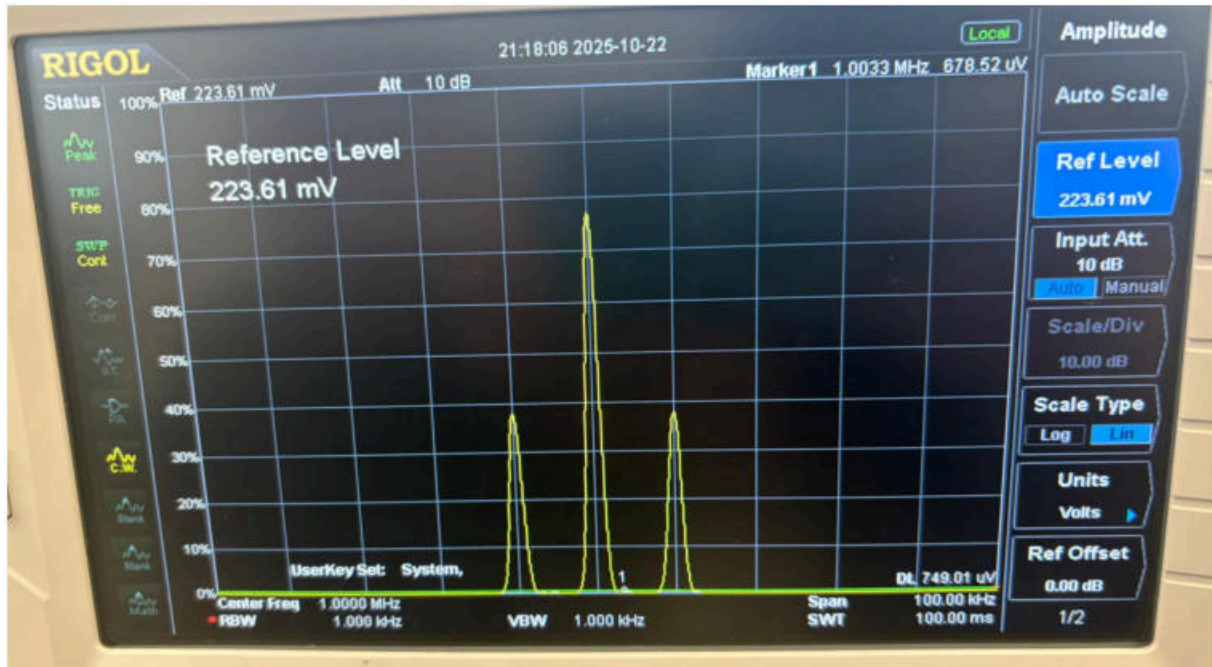
100% modulation:



IMG 20.



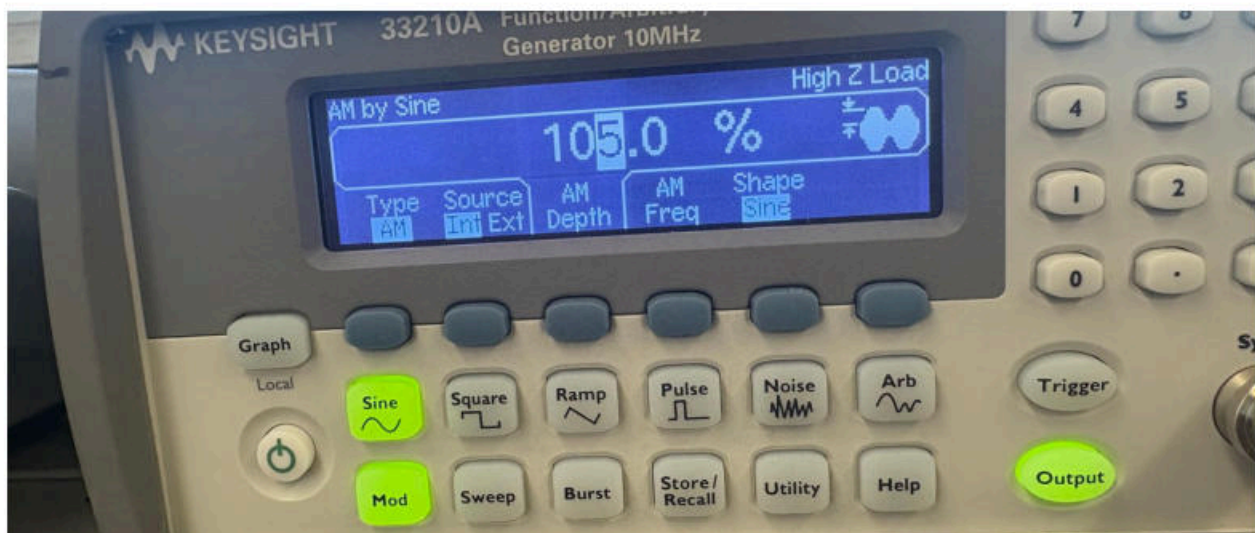
IMG 21. time domain



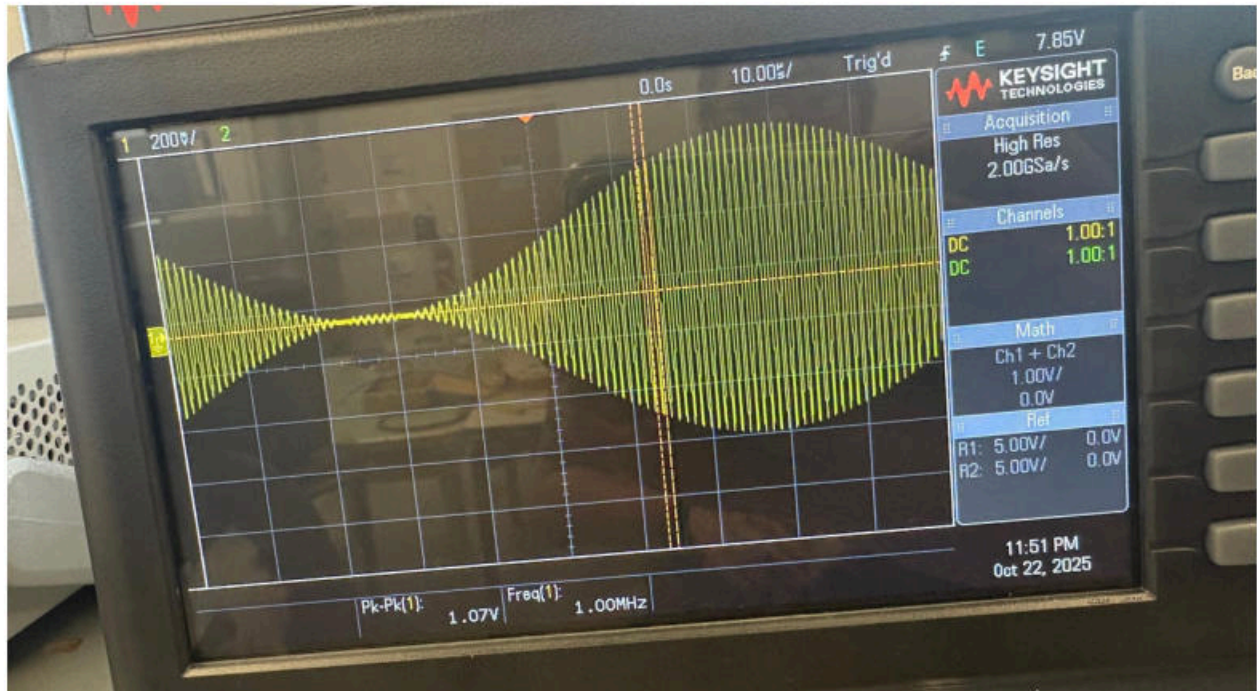
IMG 22. Frequency domain
Close to 1V

2. Increase the ma value beyond 100% and record the time and frequency data.

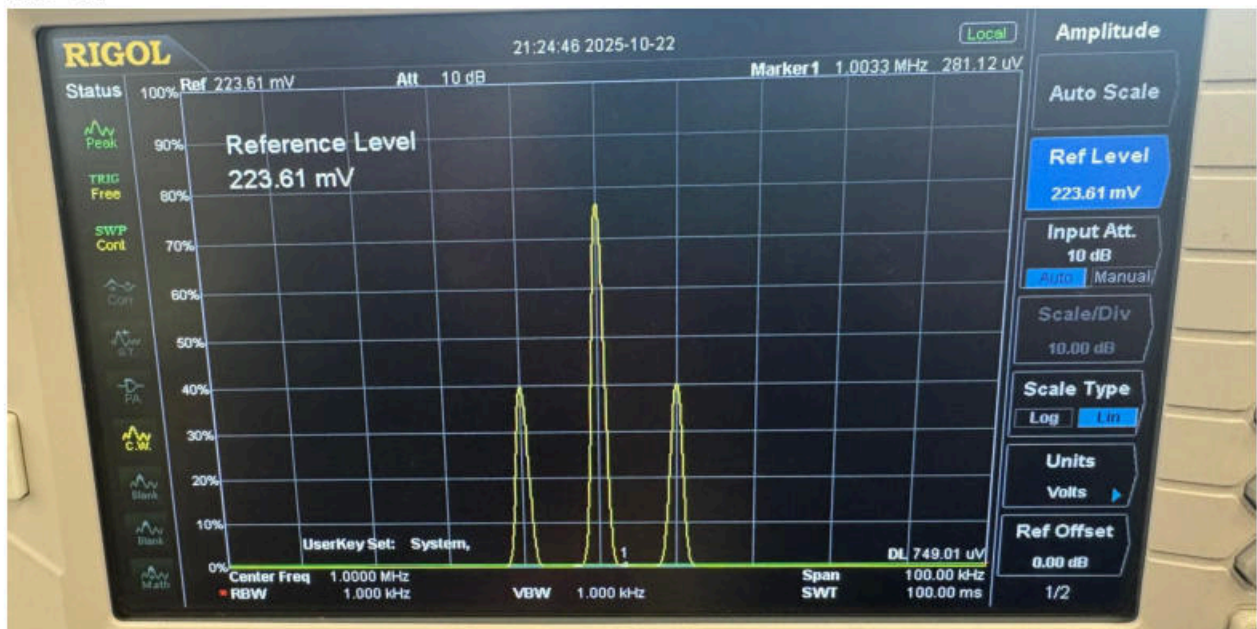
105% modulation:



IMG 23

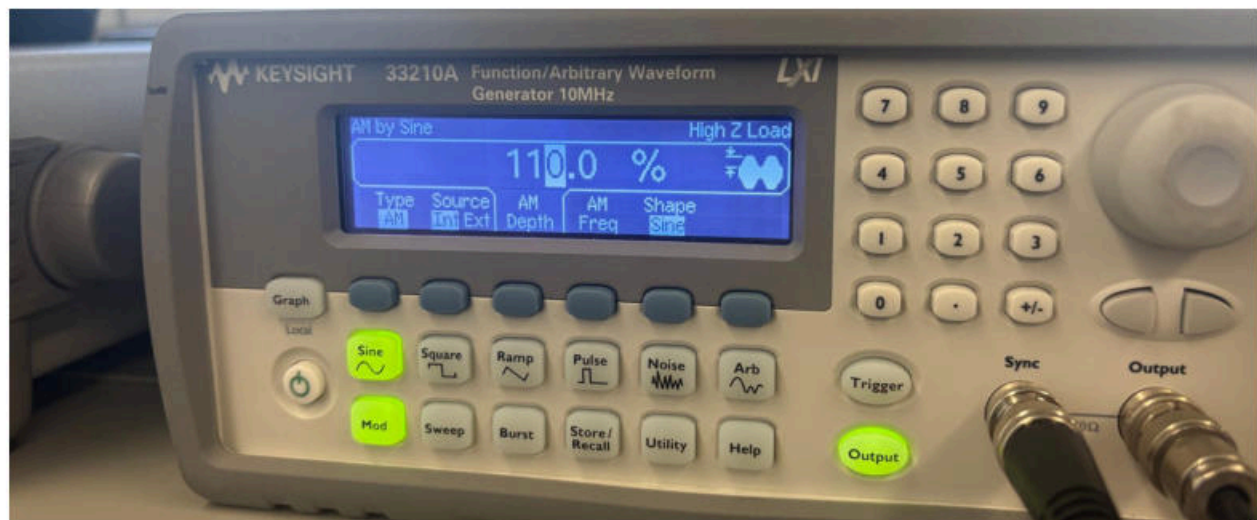


IMG 24

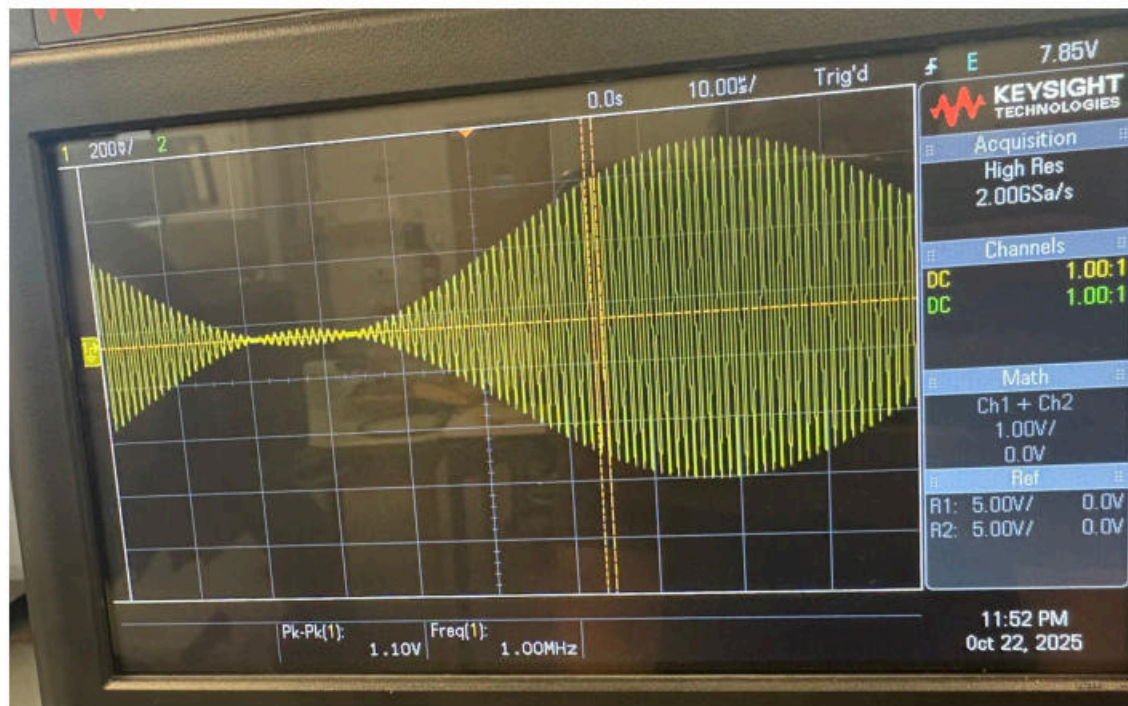


IMG 25

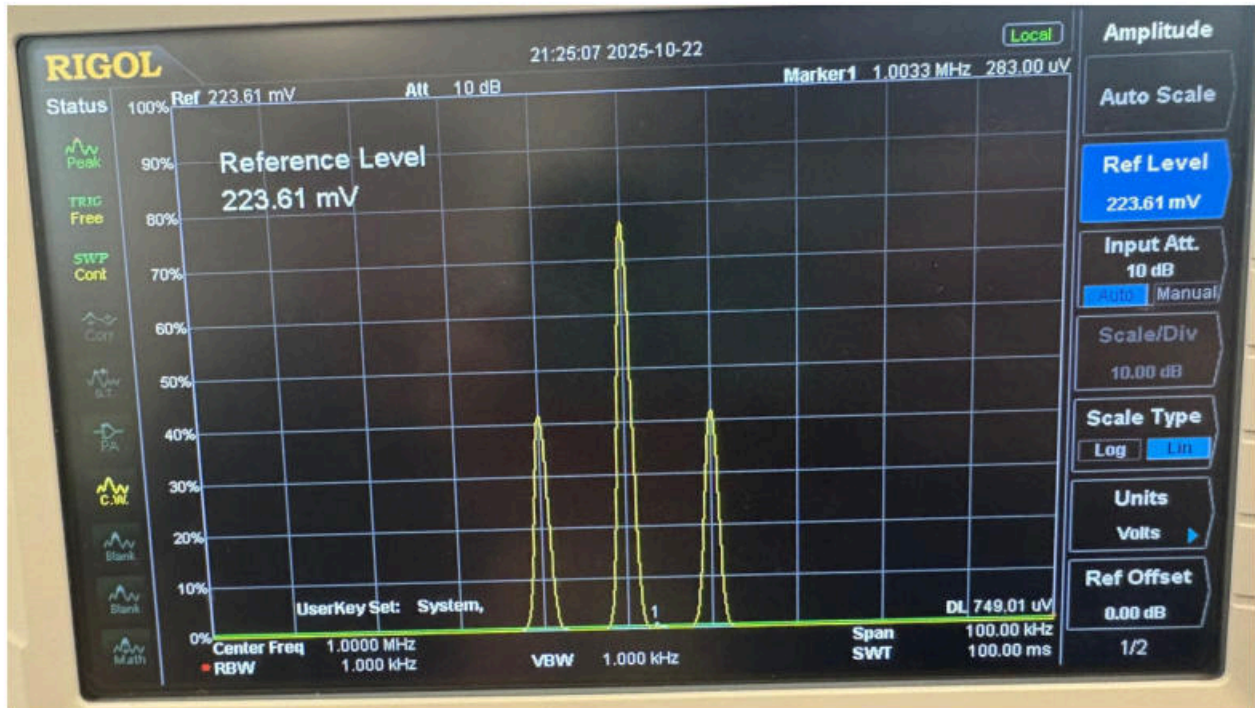
110% modulation:



IMG 26

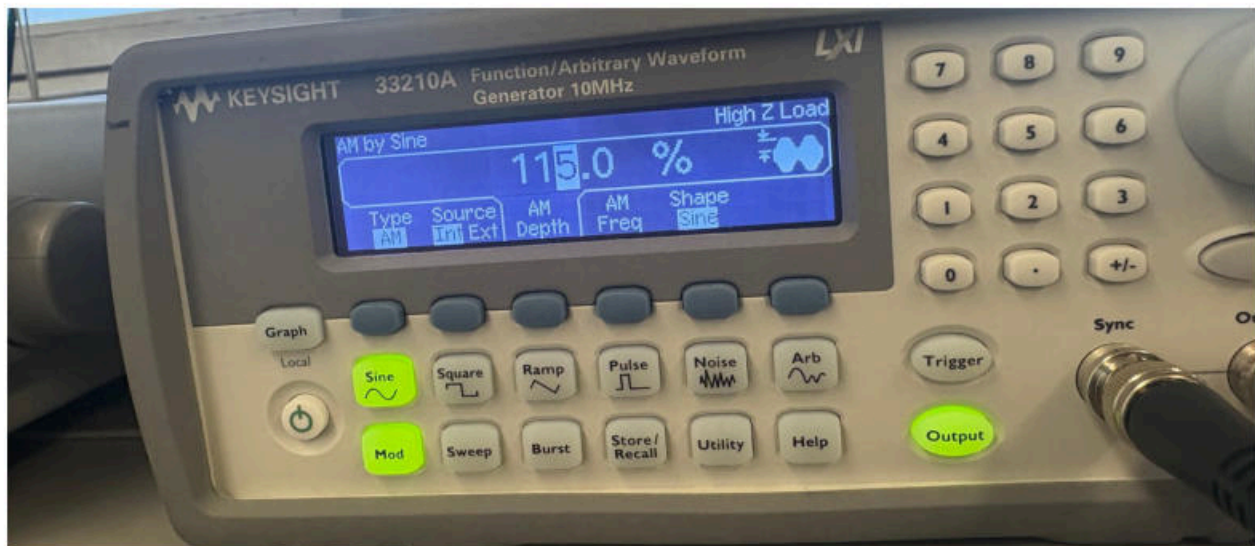


IMG 27

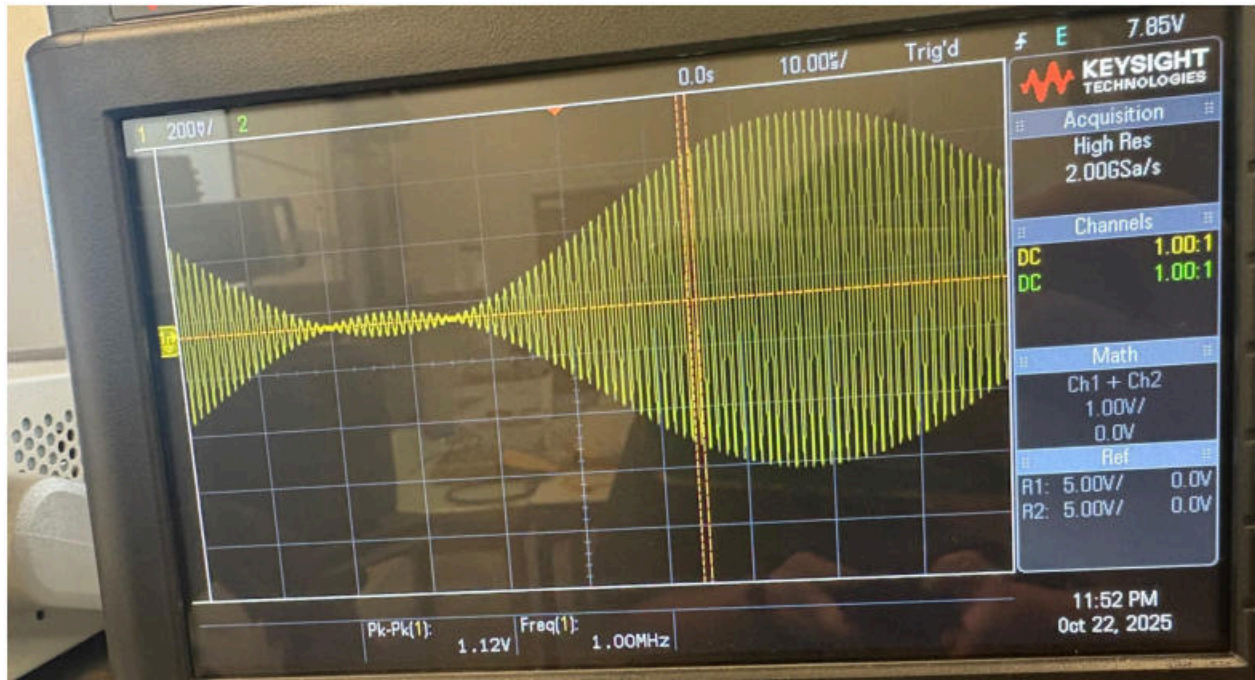


IMG 28

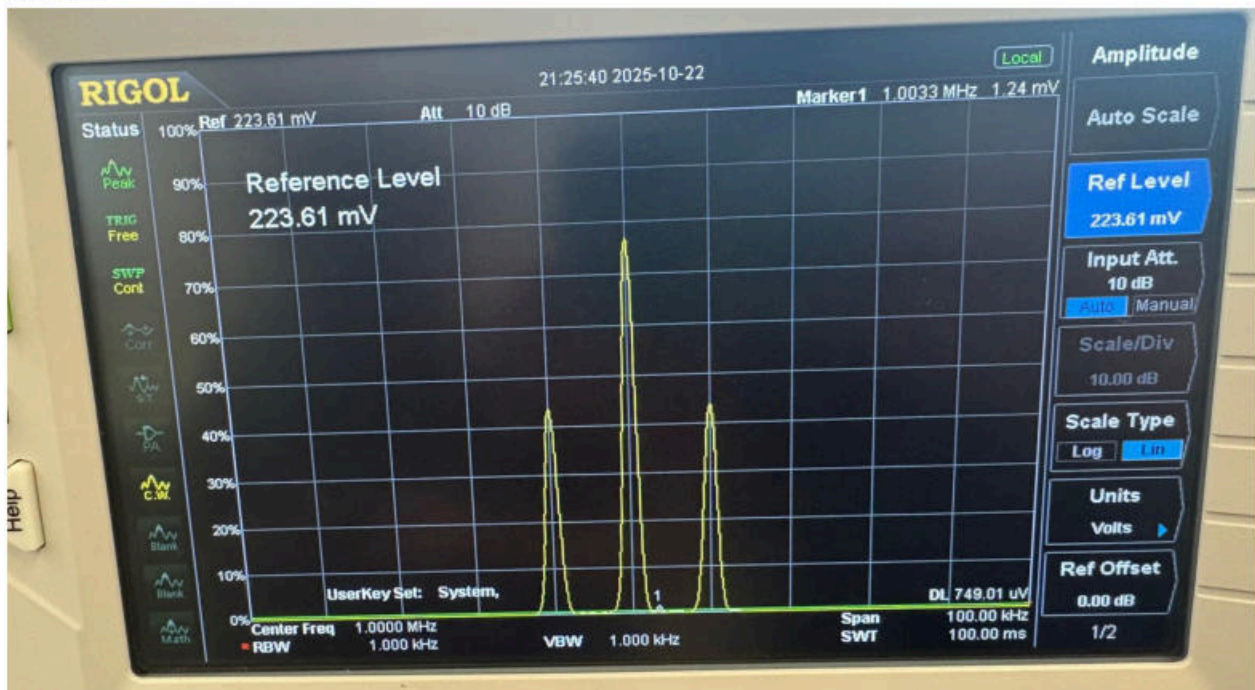
115% modulation:



IMG 29

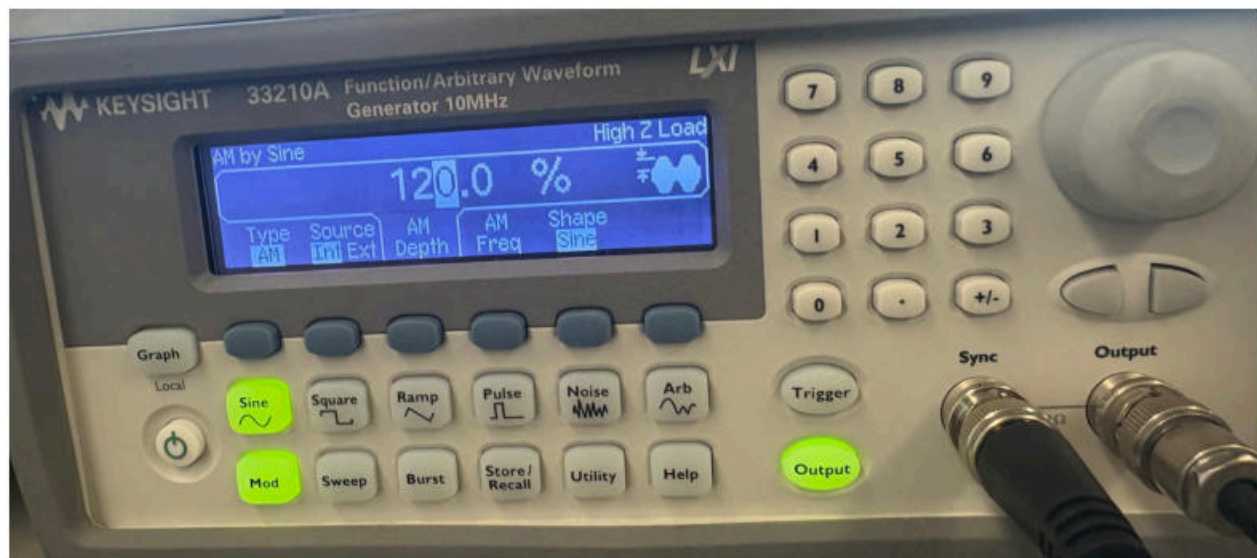


IMG 30

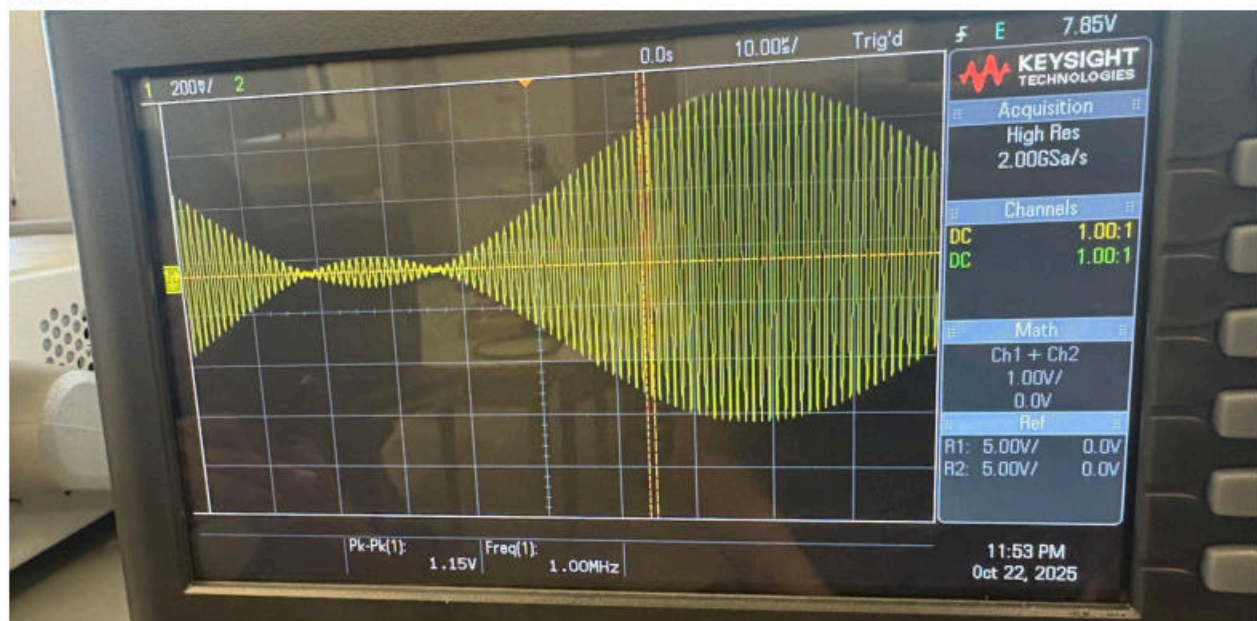


IMG 31

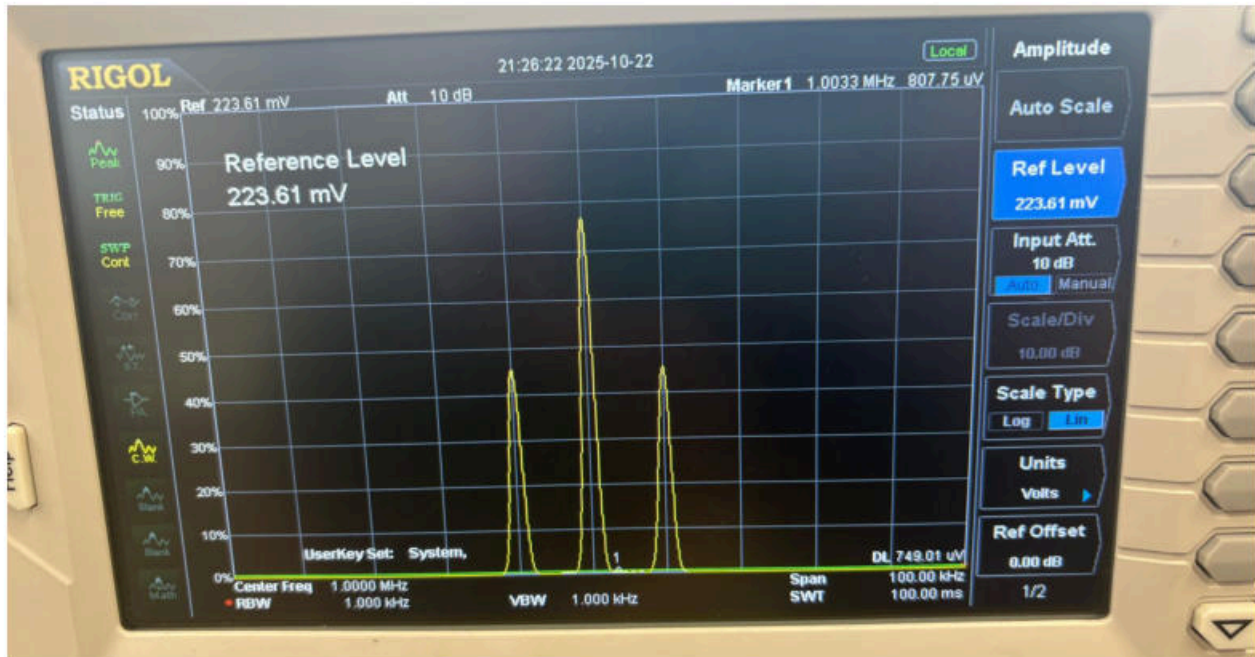
120% modulation:



IMG 32



IMG 33



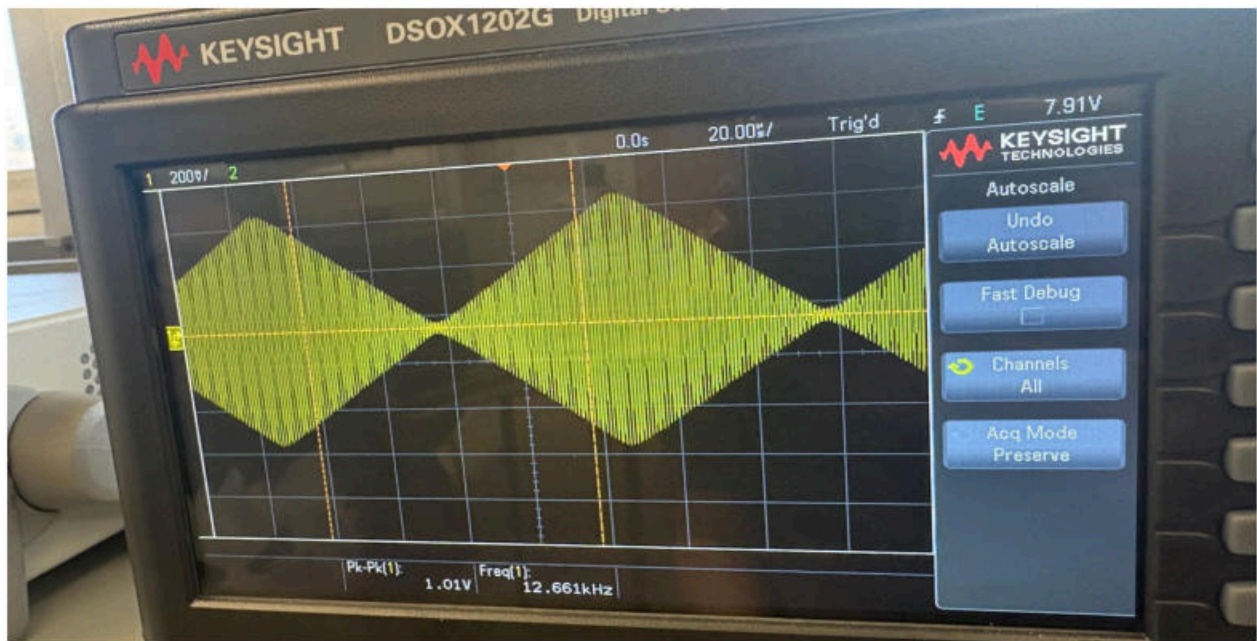
IMG 34

The modulation index cannot be set higher than 120%.

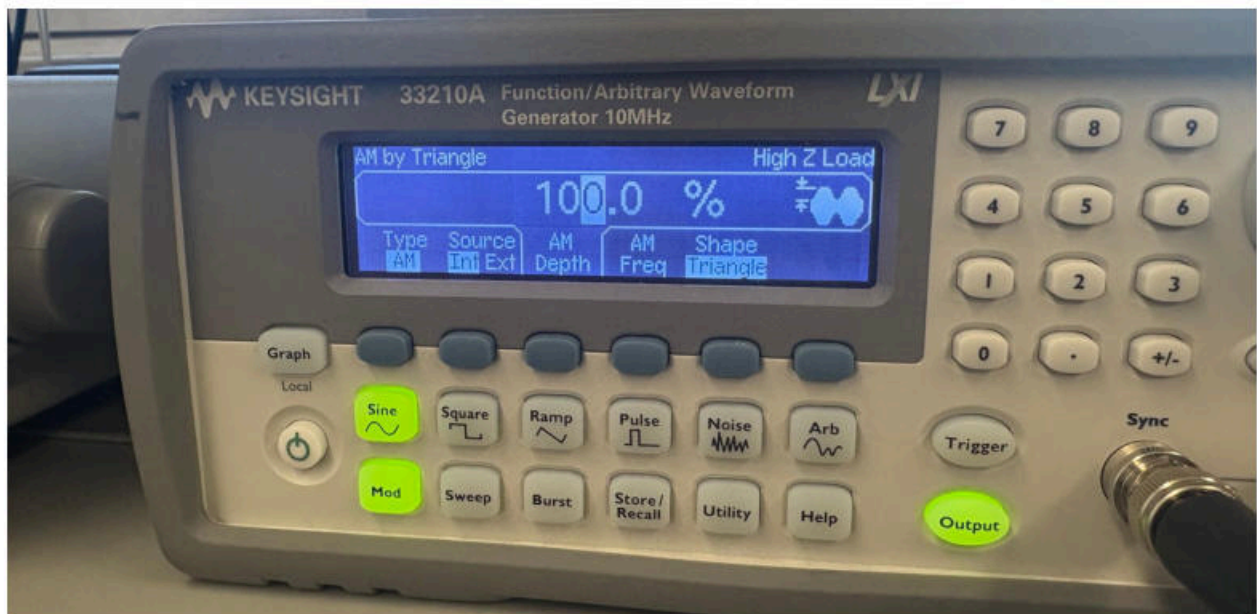
3) Using $ma = 100\%$ change the modulation signal to a triangle wave record the time and frequency data

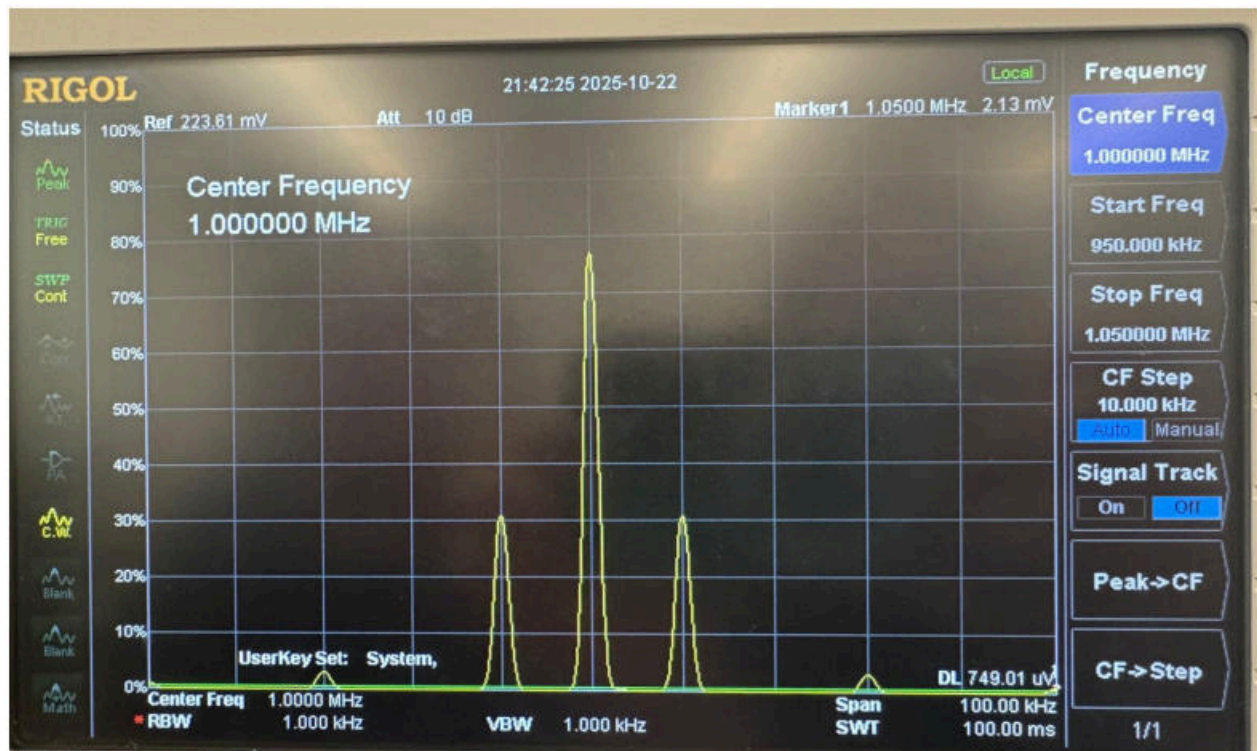
For Part 3 of the lab the modulation index was set to 100 percent and the modulating signal was changed to a triangle wave. The carrier signal remained a sine wave at all times, meaning the carrier shape itself did not change. Instead, the shape of the modulation signal controlled the shape of the envelope. By using a triangle wave as the modulating signal, the envelope of the AM waveform followed a triangular shape. This demonstrates that in amplitude modulation, changing the modulating signal changes the envelope shape, not the shape of the carrier waveform itself.

IMG 36



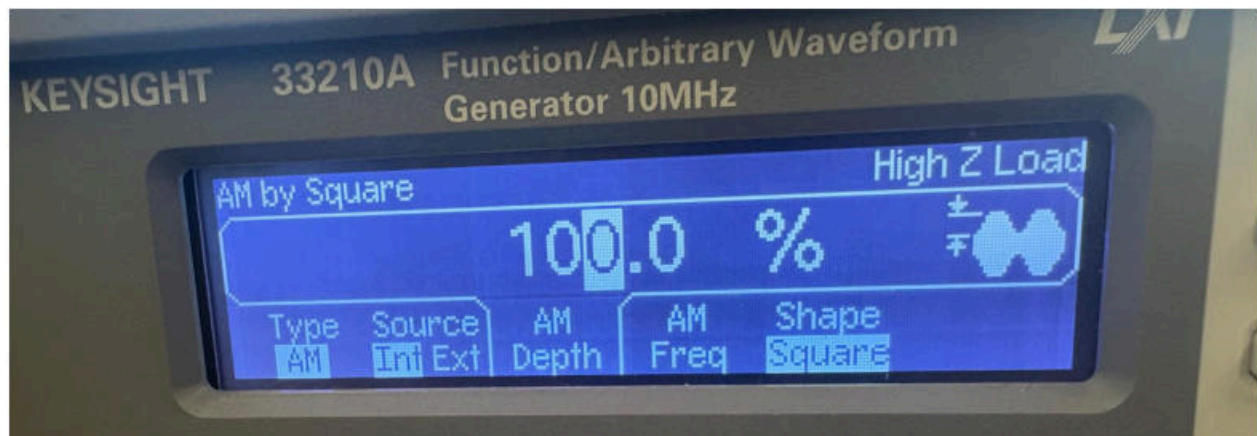
IMG 37



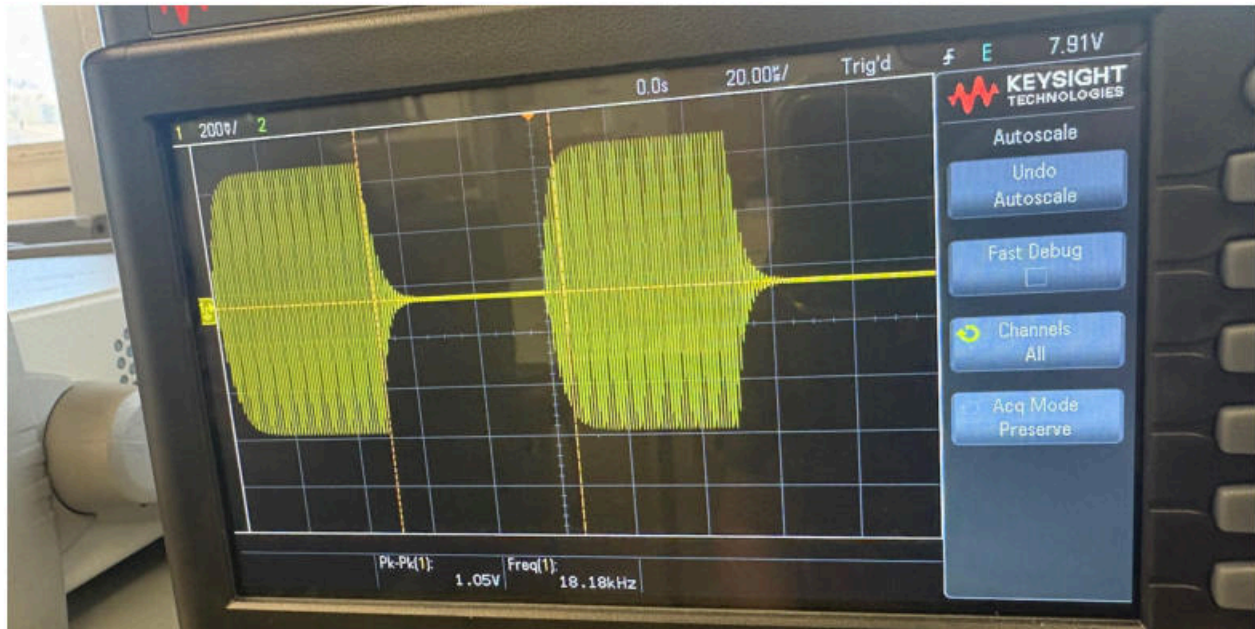


IMG 38

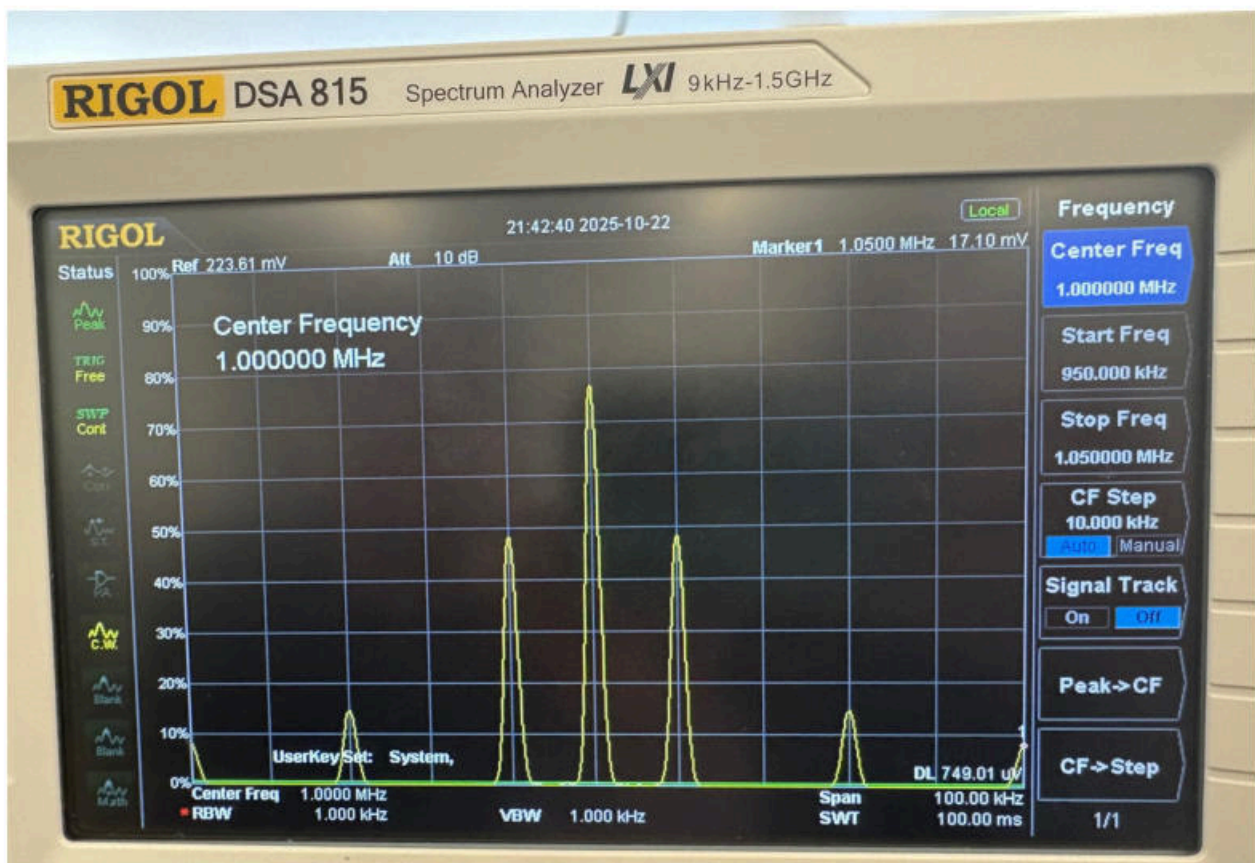
4. Using $m_a = 100\%$ change the modulation signal to a square wave record the time and frequency data



IMG 39



IMG 40



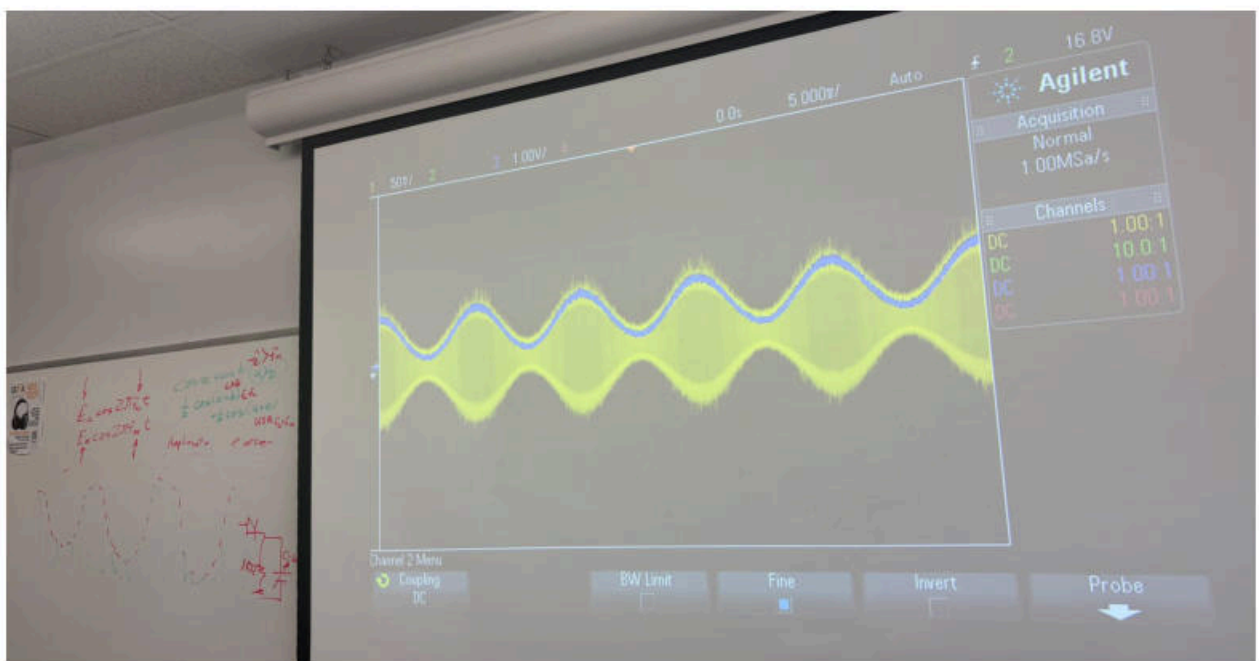
IMG 41

Parts 6 and 7 AM demodulator (maybe combined with following lab on Mixers)

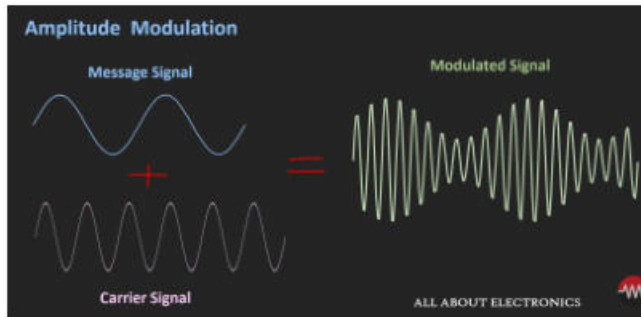
Disconnect the spectrum analyzer as only the time domain view is relevant for Demodulation.

6. Connect a sinusoidally modulated AM signal (similar signal to step 1, 50% modulation) to a diode detector circuit (1N904, R=10K to 20K), C = .001uF (use adjustable C box) and record the time domain signal (not the frequency spectrum). Adjust the capacitor value and modulation % for a good output signal. Make sure to note the Average value of the output. See the book for an equation that describes the "optimum choice" of Capacitor value, Book Equation: 5-16 shown $C \leq \frac{\sqrt{\frac{1}{m^2} - 1}}{2\pi R f_{m_{max}}}$ where $m = m_a$ equation same as from Stanford lab²
7. Increase the level of the signal in part 5 and record the same information as part 5.

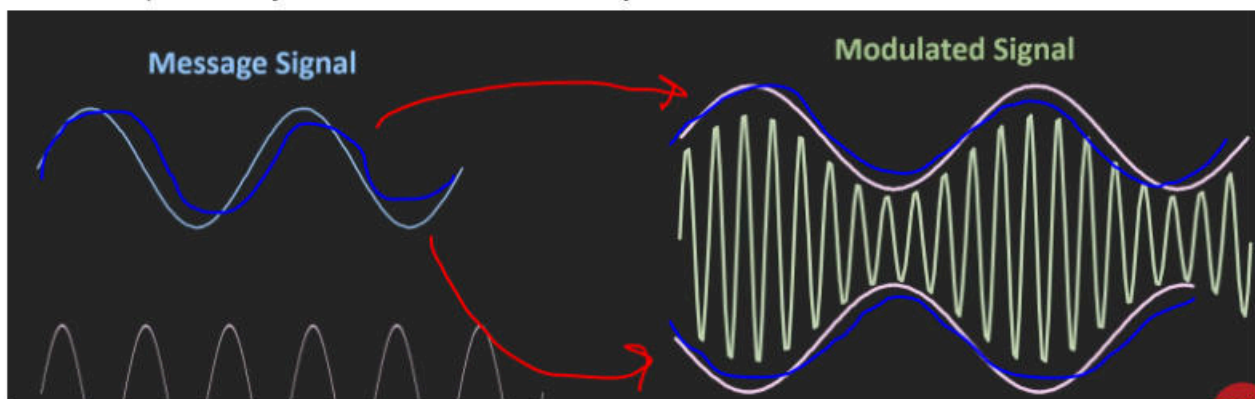
$$C \leq \frac{\sqrt{\frac{1}{m^2} - 1}}{\omega_m R}$$



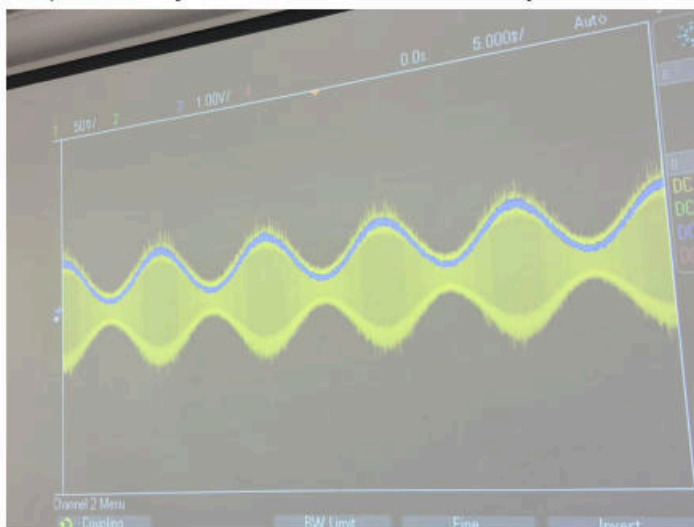
IMG 42



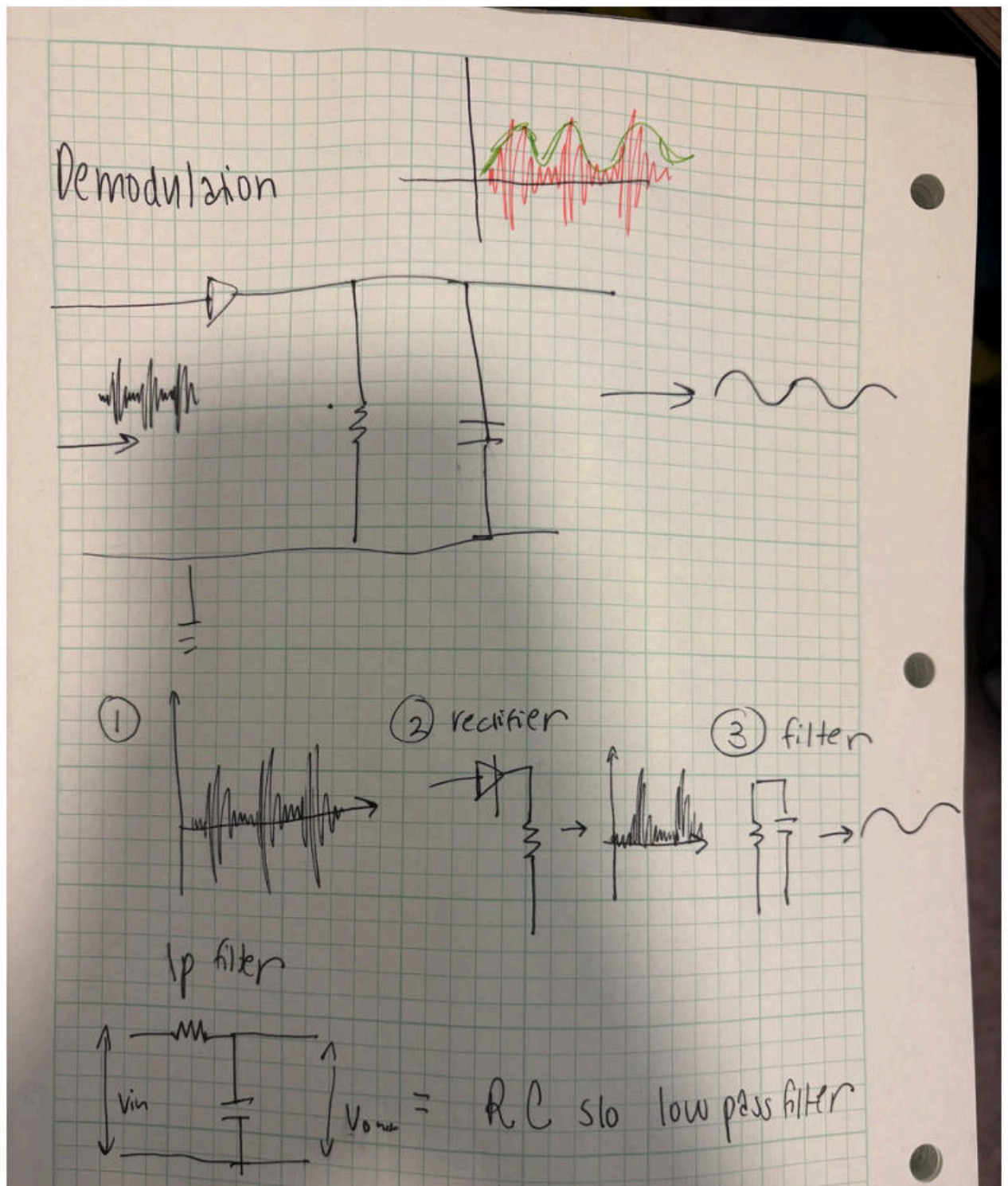
IMG 43. Modulated is the modification of amplitude frequency or phase of a sinusoidal carrier signal. AM= amplitude modulation, the amplitude of the carrier wave changes according to the message signal. The shape of the modulated signal changes based on the message signal.
Source: <https://www.youtube.com/watch?v=4JrryefRNfK>



IMG 44. Demodulation is the opposite, getting the message signal from the modulated signal .
Source: <https://www.youtube.com/watch?v=4JrryefRNfK>



IMG 45. Essentially, demodulating is getting that blue wave from the yellow modulated signal.



IMG 46. Using this circuit is how we demodulate.

For this part (part 6) of using the demodulator circuit, the voltage was measured across the capacitor. A 10 k Ω resistor was used, and the capacitor value was initially set to

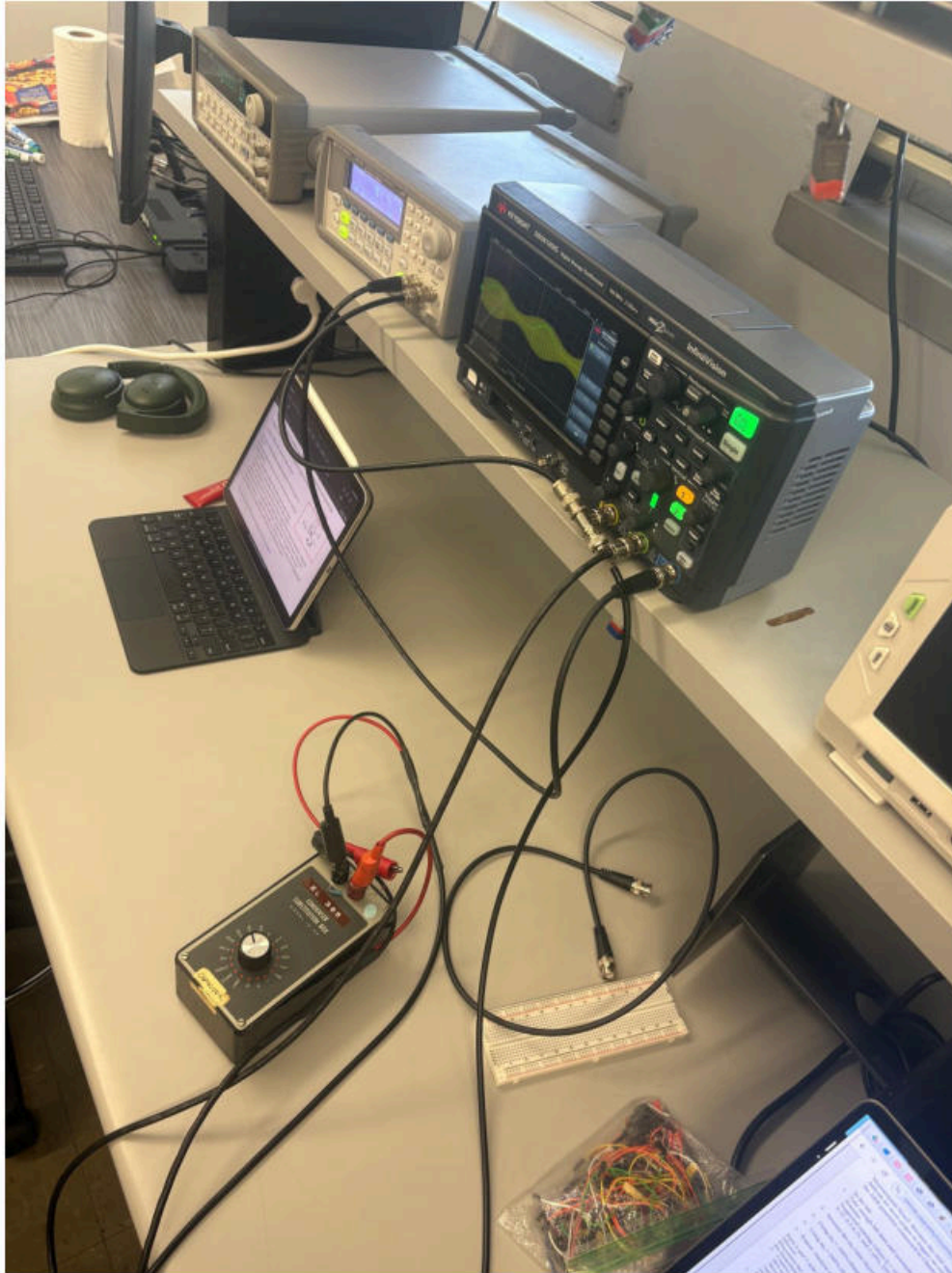
0.001 μF and then adjusted to approximately 0.0047 μF to improve the demodulated output.



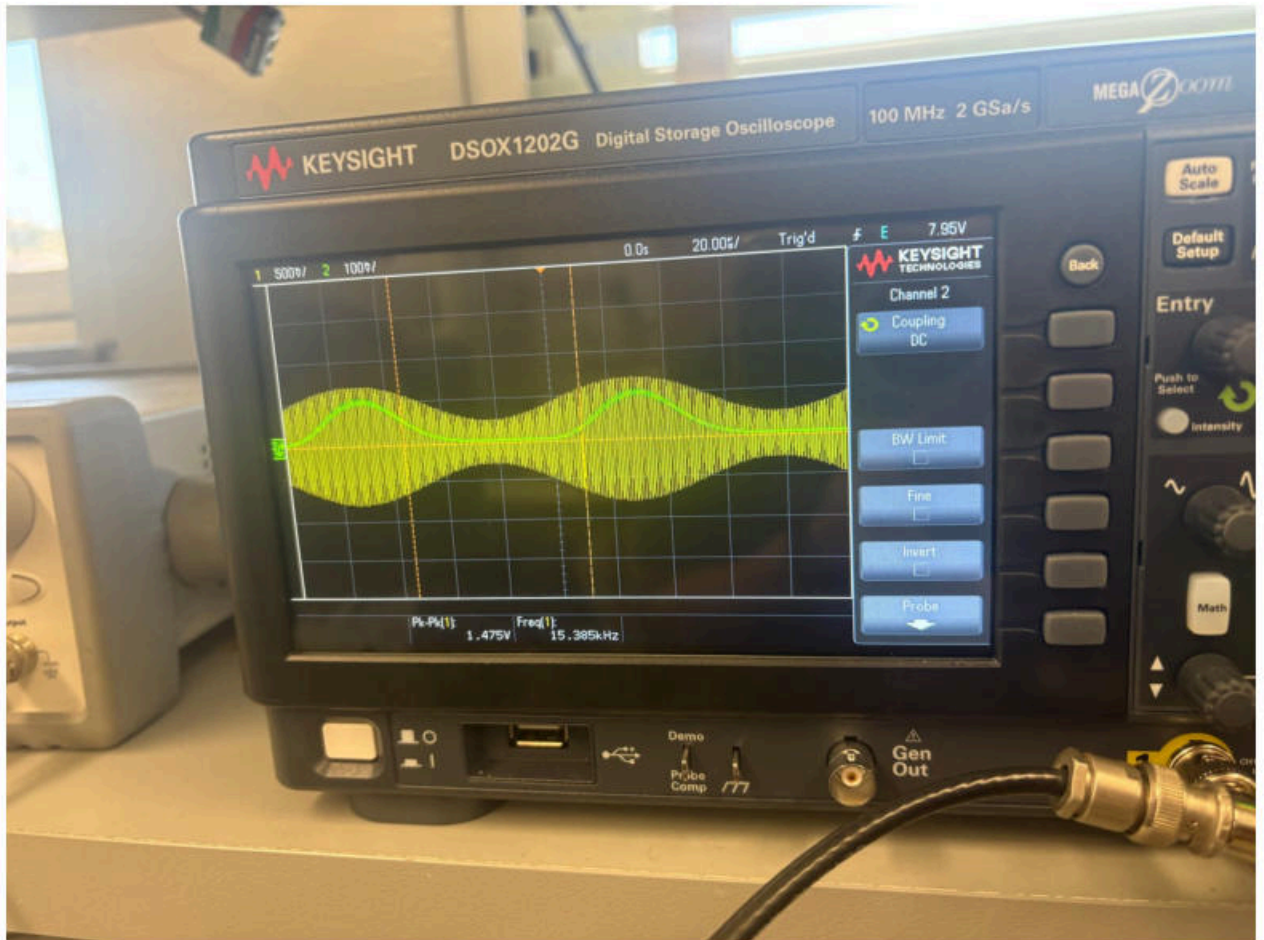
IMG 47



IMG 48. Demodulation circuit



IMG 49.



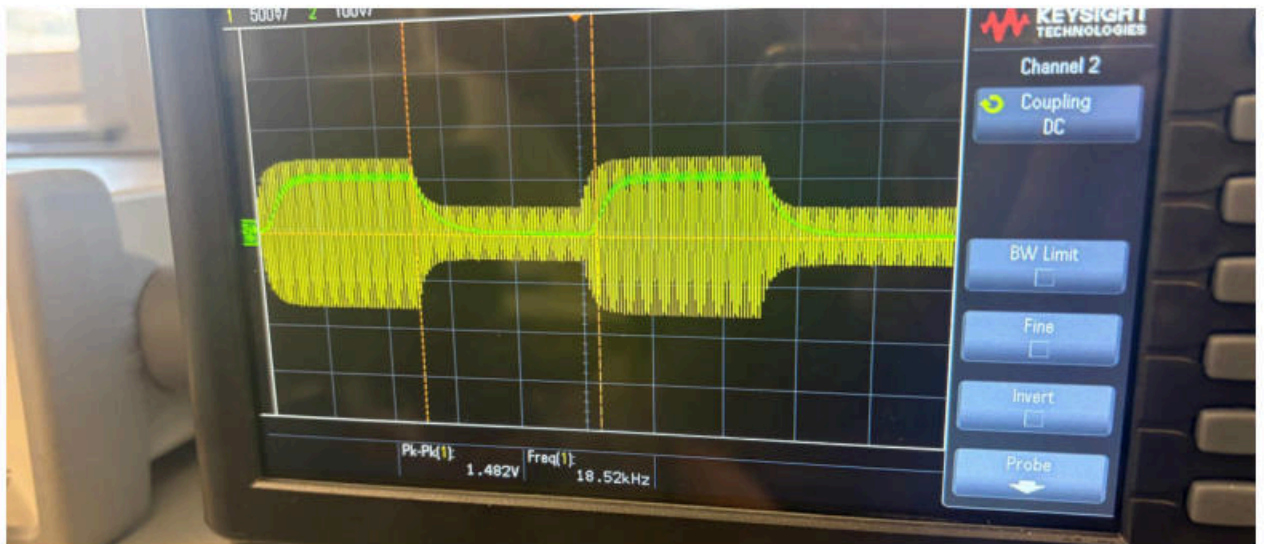
IMG 50

7. For part 7, we adjusted the capacitor (0.0047) for a better shape

Square wave

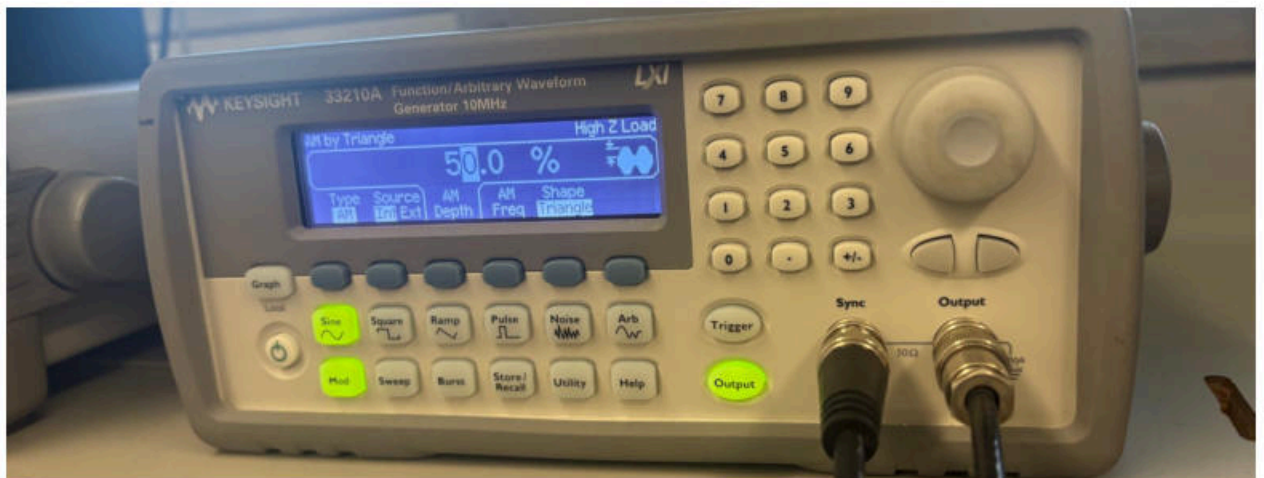


IMG 51

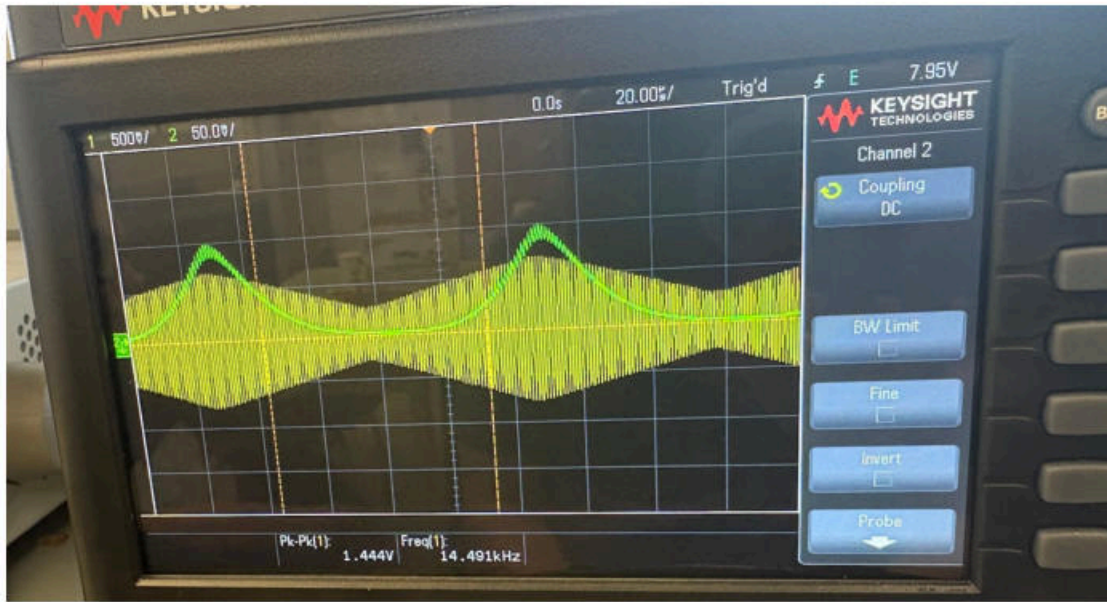


IMG 52

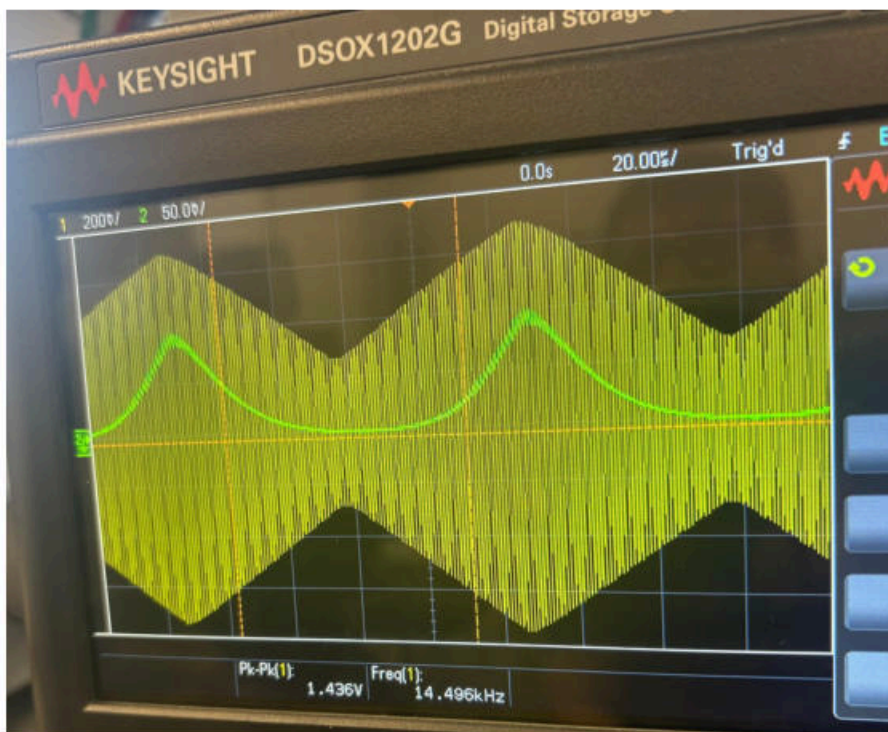
Triangle



IMG 53



IMG 54



IMG 55.

Conclusion

In Parts 6 and 7 of the lab the goal was to demodulate the AM signal (recovering the original message signal from the carrier). The yellow waveform on the oscilloscope is the AM signal, which is made up of a fast carrier wave inside a slower-changing envelope. The carrier is the high-frequency part that allows the signal to travel while the

envelope contains the actual information. The dotted line seen on the waveform is not a real signal but actually is a visual outline of the peaks of the carrier showing the shape of the envelope. When the modulation signal is changed to a triangle wave the envelope also becomes more triangular in shape, showing that the envelope shape follows the modulation signal.

To recover the original signal the carrier must be removed and only the envelope kept. This is done using a simple diode, resistor, and capacitor circuit (see image 46). The diode cuts off part of the waveform (rectification) which helps separate the envelope from the carrier. The capacitor then smooths the signal by removing the high-frequency carrier acting in part of the low-pass filter. As the capacitor value increases the output signal becomes smoother and more closely follows the envelope. Enough voltage is needed to turn on the diode, which is why signal strength matters. If the signal is over-modulated the envelope thus becomes distorted and the recovered signal will also be distorted. This lab shows why AM is popular, since the original signal can be recovered easily using a simple demodulation circuit.